SIEMENS

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SIMATIC Ident

RFID systems
XML programming for SIMATIC Ident

Application Manual

08/2025

C79000-G8976-C592-09

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We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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Introduction 1

Purpose of this documentation

This manual contains information required to program SIMATIC Ident devices via the XML interface.

Basic knowledge required

Specific knowledge of XML programming, as well as general knowledge of automation technology and identification systems, is required to understand this manual.

Trademarks

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SIMATIC ®, SIMATIC RF ®, MOBY ®, RF MANAGER ®, SIMATIC Sensors ®

Orientation in the documentation

In addition to this manual, you may need the hardware manuals of the particular Ident devices that you want to program. At this time, the following SIMATIC Ident products can be programmed via the XML interface:

- SIMATIC RF360R
- SIMATIC RF610R, RF615R, RF650R, RF680R and RF685R
- SIMATIC RF690R and RF695R
- SIMATIC RF1140R and RF1170R
- SIMATIC RF128C
- SIMATIC RF166C
- SIMATIC RF185C, RF186C, RF188C, RF186CI and RF188CI

You can find the current versions of the relevant manuals on the pages of the Siemens Industry Online Support (https://support.industry.siemens.com/cs/ww/en/ps/14970/man).

1.1 Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial

cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit
https://www.siemens.com/cybersecurity-industry.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed under
https://new.siemens.com/cert.

API description

2

2.1 Properties and area of application

Properties

This protocol represents simple XML communication for a point-to-point connection. The frames (commands/responses) are defined in XML syntax.

Area of application

Using the following XML commands, you can configure/program and diagnose Ident devices that are not connected to a controller via a user application in an Ethernet-based IT environment.

At this time, the following SIMATIC Ident products can be programmed via the XML interface:

- SIMATIC RF360R
- SIMATIC RF610R, RF615R, RF650R, RF680R and RF685R
- SIMATIC RF690R and RF695R
- SIMATIC RF1140R and RF1170R
- SIMATIC RF128C
- SIMATIC RF166C
- SIMATIC RF185C, RF186C, RF188C, RF186CI and RF188CI

Integration

The following graphic shows an example hardware configuration:. In addition to a PC with the user application, you need a compatible Ident device that is connected to the PC or a third-party controller via Ethernet.

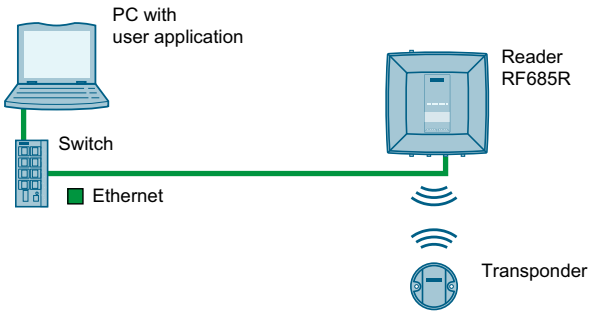


Figure 2-1 Example of a suitable hardware configuration for XML programming

2.2 Principle of operation

The XML interface is based on command/reply frames. Each command that you send to the connected Ident device via the XML programming interface is replied to by the Ident device with a reply frame regardless of whether the command was executed successfully or not. If errors occur during communication or if the command could not be executed, the reply frame contains an error description. In addition to these command/reply frames, the Ident device can also send asynchronous reports to the XML programming device.

To ensure unique assignment of the commands and replies, each command must include a unique ID. This ID is repeated in the corresponding reply frame.

Normally, the Ident device replies to a command within 5 seconds. We recommend that you program your application so that if this time is exceeded, the application starts appropriate error handling. Some commands (e.g. "setConfiguration" or "readTagIds") can take longer than 5 seconds. These commands already include information to this effect.

With the "Pending tag commands" mode, the set maximum wait time must also be considered in addition to the normal command execution time. In this mode, commands for transponder processing remain pending on the Ident device until either a transponder is detected in the antenna field or the maximum wait time has elapsed. Only then is the command answered by the Ident device.

Note

Saving and working through the commands

It is also possible to send several commands without waiting for the replies. The Ident device works through the commands in the precise order in which they were received. Note that the Ident device discards newly arriving commands if there are already approximately 100 commands waiting for execution in the device memory.

The "hostGreetings", "hostGoodbye", "heartBeat", "resetReader", "resetDevice" and "stopCommand" commands are exceptions. These are always processed immediately, regardless of whether other commands are pending for processing.

Asynchronous notifications (XML reports)

Apart from the synchronous command/reply frames, asynchronous notifications - so-called reports - are also transferred. These reports are generated by the Ident device and may require confirmation of receipt by the user application. Each transfer includes a unique ID (<id>) generated by the Ident device. In contrast to the IDs of the commands, this ID is generated by the Ident device itself. The user application can only acknowledge this notification with the same ID.

The reports are divided into events and alarm messages. An event contains data acquired by the Ident device itself. Alarm messages inform the user application automatically of special or incorrect operating conditions of the Ident device.

The reports can be transferred transacted or not transacted.

- In the non-secure mode, all reports are sent to the user application without waiting for confirmation of receipt. The report is automatically discarded if the connection to the user application does not exist or is interrupted.
- In transacted mode, the receipt of every report must be confirmed by the user application with a reply frame. If no confirmation of receipt is received within approximately 10 seconds, the Ident device sends the report to the user application again.
  If there are connection errors or interruptions, the reports are stored on the Ident device until the connection is re-established. Note that saved reports are lost when the Ident device is switched off. If necessary, activate the "Transacted transfer" function in the configuration of the Ident device if the connection between the Ident device and PC is not stable, for example due to a bad WLAN connection or if data can only be retrieved at longer intervals (batch operation).
  You will find further information on the structure of the reply frames in the section "tagEventReport (Page 133)".
  Note that the Ident device can save a maximum of 10 000 reports. If this number is exceeded, newly generated reports are discarded internally.

2.3 Demo application

A demo application created by Siemens and based on Windows .NET 4.8 is described in this section. This application serves as a model on the basis of which you can program your own user application. All XML commands listed in "Programming (Page 15)" are contained in the demo application. This allows you to test your Ident devices directly using the demo application and to run commands.

You can find the described demo application including source code files as download on the pages of "Siemens Industry Online Support (https://support.industry.siemens.com/cs/ww/de/view/109974237)".

Note

Disclaimer of liability

Note that Siemens AG accepts no liability for the demo application "RFID Reader XML Demo".

2.3.1 Startup and structure of the demo application

Requirement

.NET 4.8 Runtime is installed on the Windows PC.

Start

Follow the steps below to start the demo application on your PC:

1. Download the "SIMATIC_Ident_XML-Demoapplikation.exe" file to your PC.
   You can find the file on the pages of the Siemens Industry Online Support (https://support.industry.siemens.com/cs/ww/de/view/109974237).
2. Run the file.
   It is a self-extracting ZIP file.
3. Start the application by double-clicking on the file "RFReader.TestApp.exe".

Result: The demo application starts.

Structure

The demo application consists of the following main components:

- Demo XML API "RFReader.XmlApi.dll"
  The Demo XMP API is implemented using the DLL file. The demo application is based on the XML API. The XML API makes all XML functions available via .NET. To be able to test the XML API in your own application, you need to reference the DLL file in your project.
- User interface of the"RFReader.TestApp.exe" demo application and associated files
  Start the demo application by running the "RFReader.TestApp" file. It accesses the Demo XML API (DLL file) to be able to use the XML functions.
  This application provides you with a graphical interface with which you can send predefined commands to Ident devices. The application can communicate with multiple devices in parallel. For each connected device, a new instance of the "RFReader.XmlApi" is generated with the "Device > Connect Device" command and managed by the application (with the respective API name).

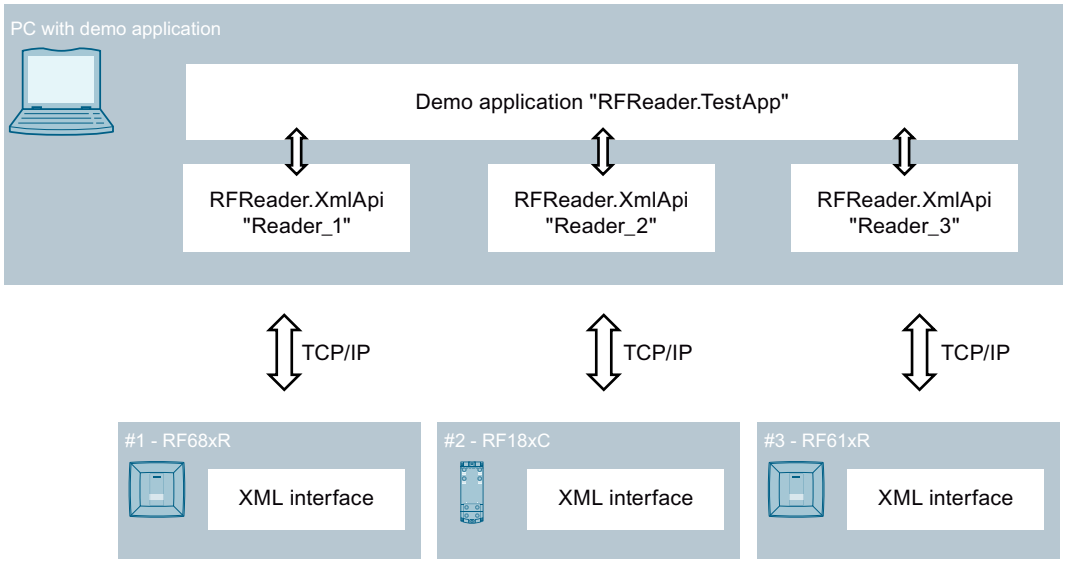
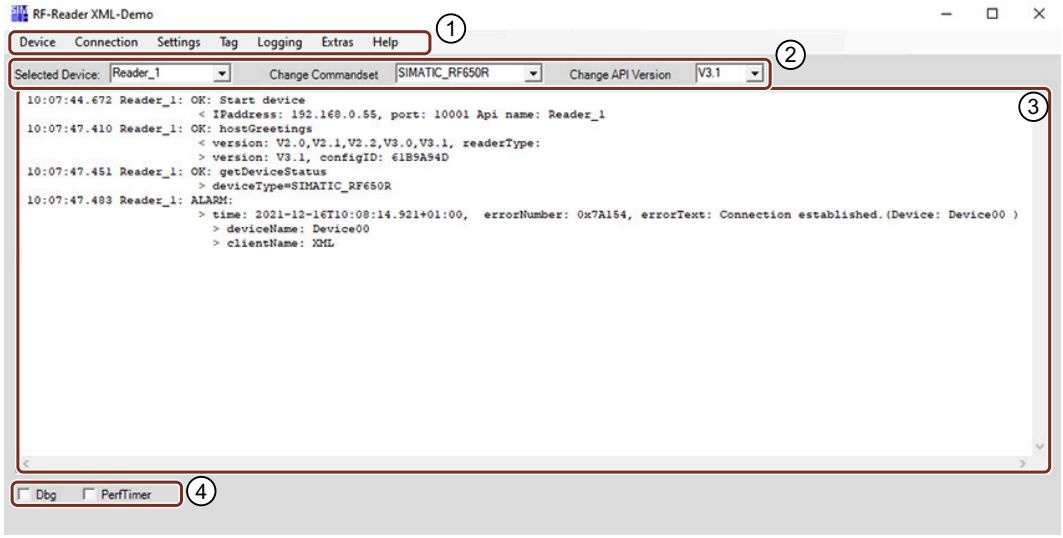


Figure 2-2 Structure/functions of the demo application

2.3.2 User interface of the demo application

The start window of the application appears after the demo application has been started. The application is divided into the following areas:



① Menu bar
② Device display
③ Log window
④ Display options

Figure 2-3 User interface of the demo application

Menu bar

The menu bar contains all available commands. These are grouped together in the relevant menus. Each individual command can be selected from the menu bar.

Table 2-1 Menu bar

Menu		Description
Device		Functions for connecting to the Ident device, disconnecting from the Ident device and for selecting devices.
XML commands		
	Connection	XML commands with which you can control the connection between the user application and Ident device.
	Settings	XML commands with which you can control the configuration of the Ident device.
	Tag	XML commands with which you can control processing of the transponder data.
Logging		XML commands for reading out and deleting log files.

Menu	Description
Extras	Functions for resetting the log window and for saving transponder data. Using the menu command "SaveTagEventReports", you can save acquired transponder data on your PC as a *.csv file or in an SQL database.
Help	Information on the demo application (e.g. XML API version)

Device display

Display of the currently selected Ident device with which you are communicating. The selection of the command scope depends on the connected device.

Log window

Text box in which all executed commands and their return values are displayed. Alarm messages and events sent by the Ident device are also displayed in the log window. The log window can be cleared with the "Extras > Clear log" menu entry.

Display options

This area contains two check boxes:

- Dbg
  With this check box, all transferred data is also displayed in XML format.
- PerfTimer
  With this check box, you can display the execution time of each command.
  The displayed time relates to the period between sending the command and the arrival of the reply on the PC.

2.3.3 Working with the demo application

Requirements

- The Ident device is connected and has started up.
- A unique IP address has been assigned to the Ident device.
- XML communication has been activated in the WBM of the Ident device.

Procedure

Follow the steps below to establish a connection to the Ident device:

1. Start the demo application.
2. Select the menu command "Device > Connect device".
3. Enter the IP address of the device in the "Device IP Address" input box.
4. If necessary, select the "Transacted" check box to enable transfer with transaction security in the application.
   Please note that transacted transfer only works if it has been activated in the WBM of the Ident device.

5. If necessary, change the API name to have the option of switching over between Ident devices when working with several device.
6. Confirm your entry with "OK".
7. Select the menu command "Connection > HostGreetings".
8. In the "Device type" input box, enter the device type with which the application should connect.
   Notation: "SIMATIC_RF6xxR" (e.g. "SIMATIC_RF680R")
   If this input box is not completed, the application connects to every connected, compatible device.
9. Confirm your entry with "OK".

Remember that after connecting to the Ident device, the "HostGreetings" command must always be executed first.

The connection to the Ident device is established. The API name of the currently active device is displayed in the "Selected device" text box. All commands are sent exclusively to this device.

You can communicate with several Ident devices at the same time. To communicate with another device, repeat the actions described above. Once you have established a connection to multiple devices, you can switch conveniently from one to the other using the "Ident device > Select device" menu command.

After you have established the connection to an Ident device and have executed the "HostGreetings" command, you can communicate with the device. To do this, various commands are available in the menus. These commands are described in the following sections.

2.4 Compatibility

The XML API of the demo application was released for the first time with version V2.0. Since then, the XML API has been expanded by some functions and commands and many additional Ident devices and product families have been added that can be programmed using the XML API.

The XML API versions are backward-compatible. The following table provides an overview of the compatible devices and versions.

Note

Umbrella terms for individual products

Note that umbrella terms are used for individual products in the rest of the documentation. This relates to all readers of the RF600 product family, the communications modules as well as the readers of the RF1100 product family. The following umbrella terms are used:

- The name "RF61xR/RF650R/RF68xR" stands for all RF610R, RF615R, RF650R, RF680R and RF685R readers.
- The name "RF69xR" stands for all RF690R and RF695R readers.
- The name "RF600" stands for all readers of the RF600 product family.
- The name "RF1xxC" stands for all RF166C communications modules as well as RF185C, RF186C, RF188C, RF186CI and RF188CI.
- The name "RF1100" stands for all RF1140R and RF1170R readers.

Table 2-2 Overview of compatible versions

Device/Product family	Firmware version of the device / product family	Protocol version of the XML API (XML API version)
RF61xR/RF650R/RF68xR	V1.0 ... V2.0	V2.0
	V2.1 ... V2.2	V2.0; V2.1
	V2.2.1 ... V3.4	V2.0 ... V2.2
	V4.0 ... V4.0.1	V2.0 ... V3.0
	As of V4.1	V2.0 ... V3.2
RF69xR	V1.0 or higher	V3.1; V3.2
RF1xxC	V1.3 ... V2.0	V3.0
	As of V2.1	V3.0 ... V3.2
RF12xC	V1.0	V3.1; V3.2
RF360R	V1.0 ...V2.0	V3.0
	As of V2.1	V3.0 ... V3.2
RF1100	V1.0 or higher	V3.2

Programming 3

3.1 Command execution and responses

This section describes all XML commands and responses that you can send to or receive from an Ident device via the XML API. All XML commands and responses are encoded according to UTF-8.

Each command sent by the XML API application is replied to by the Ident device with a reply frame. If the command was executed successfully, the reply frame has the value "0" in the "ResultCode" parameter. If other values are returned in this parameter, this means that the command was not executed successfully. In this case, the returned value corresponds to the error code.

Note

Parameter sequence in commands

In all commands (and responses), note that the parameters within a hierarchy level can be in any sequence.

Note

Number of parameters depends on the device type and the API/firmware version

Note that further optional parameters may be added in later development stages. Checking the number of parameters could therefore lead to incompatibilities.

3.2 Command/response overview

The XML API of the demo application was released for the first time with version V2.0. Since then, the XML API has been expanded by some functions and commands and many additional Ident devices and product families have been added that can be programmed using the XML API.

The following tables provide you with an overview of all XML commands and responses stored in the XML API. There is a separate table for each command/response category. Based on the tables, you can determine the XML API protocol version as of which the respective commands are contained in the XML API and the device firmware versions with which they are compatible. In addition, each XML command/XML response section contains a separate

overview of the XML API protocol version (API version) and device firmware versions as of which the respective command or response is valid.

Note

Umbrella terms for individual products

Note that umbrella terms are used for individual products in the rest of the documentation. This relates to all readers of the RF600 product family, the communications modules as well as the readers of the RF1100 product family. The following umbrella terms are used:

- The name "RF61xR/RF650R/RF68xR" stands for all RF610R, RF615R, RF650R, RF680R and RF685R readers.
- The name "RF69xR" stands for all RF690R and RF695R readers.
- The name "RF600" stands for all readers of the RF600 product family.
- The name "RF1xxC" stands for all RF166C communications modules as well as RF185C, RF186C, RF188C, RF186CI and RF188CI.
- The name "RF1100" stands for all RF1140R and RF1170R readers.

XML commands

Table 3-1 Overview of the "Connections" XML commands and their version dependencies

XML command	Description	As of XML API version	From firmware version					
			RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
hostGreetings	Establish connection.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
hostGoodbye	Terminate connection.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
heartBeat	Monitor connection.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
setIPConfig	Specify Ethernet address of device.	V1.0	V2.2.1	V1.0	V1.3	V1.0	V1.0	V1.0
getIPConfig	Read out Ethernet address of device.	V1.0	V2.2.1	V1.0	V1.3	V1.0	V1.0	V1.0

Table 3-2 Overview of the "Device settings" XML commands and their version dependencies

XML command	Description	As of XML API version	From firmware version					
			RF61xR/ RF650R/ RF68xR	RF69 xR	RF1xxC	RF12xC	RF360R	RF1100
prepare FwUpdate	Starts the frame for the firmware update of the device.	V3.1	V4.1	V1.0	V2.1	V1.0	V2.1	V1.0
transferFW	Transfers the firmware of the device.	V3.1	V4.1	V1.0	V2.1	V1.0	V2.1	V1.0
executeFW	Activates the transferred firmware in the device.	V3.1	V4.1	V1.0	V2.1	V1.0	V2.1	V1.0

XML command	Description	As of XML API version	From firmware version					
			RF61xR/ RF650R/ RF68xR	RF69 xR	RF1xxC	RF12xC	RF360R	RF1100
set Configuration	Transfers a configuration to the device and saves it retentively.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
get Configuration	Reads out the saved configuration from the device.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
get ConfigVersion	Reads out the version ID of the saved configuration from the device.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
getActive Configuration	Reads out the current values from the device.	V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	--
getLogfile	Reads out the logbook of the device.	V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
resetLogfile	Deletes the logbook of the device.	V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
setLogEntry	Sending log information to the device.	V3.2	V4.3	V1.2	V2.3	V1.1	V2.3	V1.2
setParameter	Sets a value of a device parameter.	V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
getParameter	Reads out a device parameter.	V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
set ParameterList	Sets values of multiple device parameters.	V3.0	V4.0	V1.0	V2.0	V1.0	V2.0	V1.0
get ParameterList	Reads out multiple device parameters.	V3.0	V4.0	V1.0	V2.0	V1.0	V2.0	V1.0
setIMParameter	Sets an I&M parameter.	V3.1	--	V1.0	--	V1.0	--	V1.0
getIMParameter	Reads out an I&M parameter.	V3.1	--	V1.0	--	V1.0	--	V1.0
setTime	Sets the internal device clock.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
getTime	Reads the time of day of the internal device clock.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
setIO	Sets the digital outputs of the device.	V1.0	V1.0	V1.0	V1.3	--	--	--
getIO	Reads the states of the digital inputs/outputs of the device.	V1.0	V1.0	V1.0	V1.3	--	--	--
resetReader	Resets the device. Note: Present for compatibility reasons. This command has been replaced by the "resetDevice" command.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
resetDevice	Resets the device.	V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	V1.0
getReader Status	Reads out the status information of the device. Note: Present for compatibility reasons. This command has been replaced by the "getDeviceStatus" command.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
getDevice Status	Reads out the status information of the device.	V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	V1.0

XML command	Description	As of XML API version	From firmware version					
			RF61xR/ RF650R/ RF68xR	RF69 xR	RF1xxC	RF12xC	RF360R	RF1100
getCM ReaderStatus	Reads out the status information of the devices connected to the communications module.	V3.0	--	--	V1.3	V1.0	V1.0	--
getAllSources	Reads out the names of all configured read points of the device.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
getCLPConfig	Reads the status of the CLP interface.	V3.2	--	V1.0	--	--	--	--

Table 3-3 Overview of the "Transponder processing" XML commands and their version dependencies

XML command	Description	As of XML API version	From firmware version					
			RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
editBlacklist	Save transponders IDs to or remove them from the "Black list".	V2.0	V1.0	V1.0	--	--	--	--
getBlacklist	Reads out the transponder IDs from the "Black list".	V2.0	V1.0	V1.0	--	--	--	--
triggerSource	A read point is displayed for reading. Detected transponders are reported by "tagEventReport".	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
readTagIDs	The read point performs the desired number of inventories and returns the detected transponders in the reply frame.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
writeTagID	Writes a new ID to the transponder.	V1.0	V1.0	V1.0	--	--	--	--
getObserved TagIDs	Returns all transponders that have the "observed" status.	V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
readTag Memory	Reads out data from a memory area of a transponder.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
writeTag Memory	Writes data to a memory area of a transponder.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
readTagField	Reads out data from a symbolically addressed memory area of a transponder.	V1.0	V1.0	V1.0	V2.2	V1.0	V2.2	V1.0
writeTagField	Writes data to a symbolically addressed memory area of a transponder.	V1.0	V1.0	V1.0	V2.2	V1.0	V2.2	V1.0
killTag	Disables a transponder.	V1.0	V1.0	V1.0	--	--	--	--
lockTagBank	Locks a memory area of a transponder.	V1.0	V1.0	V1.0	--	--	--	--

XML command	Description	As of XML API version	From firmware version					
			RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
getTagStatus	Reads out the status information of the transponder.	V3.0	--	--	V1.3	V1.0	V1.0	--
stopCommand	Ends a running/pending command.	V3.0	V4.0	V1.0	V2.0	V1.0	V2.0	V1.0

XML reports

Table 3-4 Overview of XML reports and their version dependencies

XML report	Description	As of XML API version	From firmware version					
			RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
tagEventReport	Sends the transponder data of the respective transponder on every status change (glimpsed, observed, lost).	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0
rssiEventReport	Signals every change in the RSSI values when reading the transponders.	V1.0	V1.0	V1.0	--	--	--	--
ioEventReport	Signals every change of a digital input/output.	V1.0	V1.0	V1.0	V1.3	--	--	--
logEventReport	Signals new logbook entries.	V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	V1.0
presenceEvent Report	Signals the transponders currently located in the antenna field.	V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	V1.0
lastAccess EventReport	Sends the transponder data of the respective transponder on a successful transponder command.	V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	--
alarm	Status, warning or error messages of the device.	V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

3.3 XML commands/responses

Note

Displayed parameters and optional rows

The following sections explain the XML commands and responses, as well as reports in detail. Note that all parameters that can theoretically be displayed will be described. The availability of the parameters depends on the Ident device used, the firmware version or the parameter assignment in the WBM. These parameters are designated by the code "*".

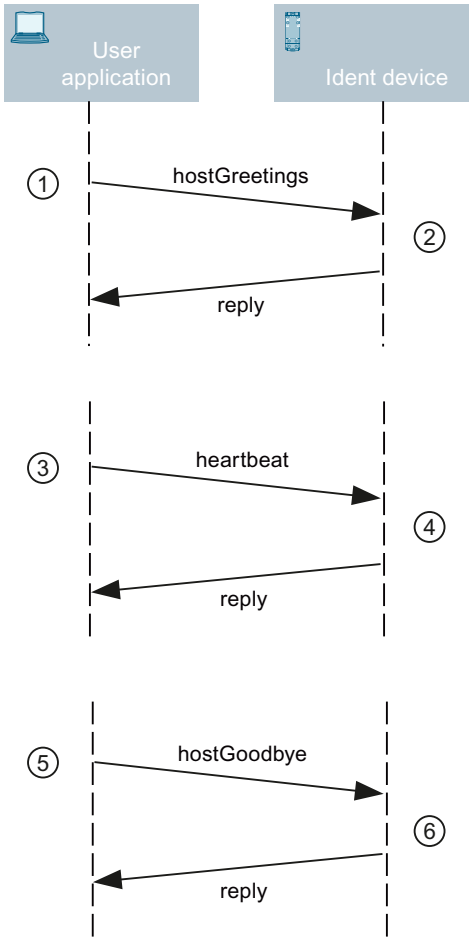
Some of the parameters are optional: These parameters are designated by the code "opt".

3.3.1 Connections

This section describes all the commands with which you can control the connection between the user application and the Ident device.

The following diagram shows how a connection is established and terminated.

Table 3-5 Sequence of connection establishment / termination

Connection establishment / termination	Step	Description
	1	The user application sends the "hostGreetings" command to the Ident device.
	2	The Ident device returns a positive reply frame.
	3	After the connection has been established, the user application communicates with the Ident device. It sends, for example, a heartbeat command at regular intervals.
	4	For each command, the Ident device sends a reply frame.
	5	The user application sends the "hostGoodbye" command to terminate the connection.
	6	The Ident device sends a positive reply frame. The Ident device then terminates the existing TCP/IP connection.

Note that the Ident device replies with the "ERROR_INVALID_READER_STATUS" error message if the commands are sent without a preceding "hostGreetings" command.

After executing the command, the device blocks connection requests of new clients for 30 seconds. By periodically sending "heartBeat" commands within a period of 30 seconds, you can make sure that no other user applications access the device.

3.3.1.1 hostGreetings

Table 3-6 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

All communication with an Ident device must start with the "hostGreetings" command. Using this command, the device detects the user application connected to the XML interface and its version. In turn, the firmware version of the device is stored in the reply frame of the device. In this way, the XML API can determine which commands are supported by the connected device.

Note that the Ident device replies with the "ERROR_INVALID_READER_STATUS" error message if the commands are sent without a preceding "hostGreetings" command.

Note that up to 20 seconds can elapse until a reply frame is received.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <hostGreetings>
        <readerType> value readerType </readerType>     //opt
        <supportedVersions>
            <version> value_version </version>
            <version> value_version </version>         //opt
            ...
        </supportedVersions>
    </hostGreetings>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <hostGreetings>
        <returnValue>
            <version> value_version </version>
            <configID> value_configID </configID>
        </returnValue>
    </hostGreetings>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ readerType	Fixed values	SIMATIC_RF610R SIMATIC_RF615R SIMATIC_RF650R SIMATIC_RF680R SIMATIC_RF685R SIMATIC_RF690R SIMATIC_RF695R SIMATIC_RF166C SIMATIC_RF185C SIMATIC_RF186C SIMATIC_RF188C SIMATIC_RF186CI SIMATIC_RF188CI SIMATIC_RF128C SIMATIC_RF360R SIMATIC_RF1140R SIMATIC_RF1170R	Device type If the connected device does not match the value specified, "ERROR_PARAMETER_ILLEGAL_VALUE" is returned. If this parameter is not specified, the connected device type is not checked.
value_version	Alphanumeric text	Vx.x	Supported XML API protocol version (See section "Compatibility (Page 13)")
value_configID	Alphanumeric text	--	Unique identifier of the transferred configuration. The ID can also be read using the "getConfigVersion" function.

3.3.1.2 hostGoodbye

Table 3-7 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

This command ends communication with the Ident device and terminates the TCP/IP connection.

In the default setting, the device continues working with its current settings. This allows the device to work independently. The data accumulating when the reader is operating independently is stored in the buffer. You will find further information on the buffer or on asynchronous notifications in the section "Principle of operation (Page 8)".

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <hostGoodbye>
    </hostGoodbye>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <hostGoodbye/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier

3.3.1.3 heartBeat

Table 3-8 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, it is possible to check whether the connection between XML API and the connected Ident device has been interrupted (e.g. wire break) or whether the device is out of operation (e.g. network failure).

After executing the command, the device blocks connection requests of new clients for 30 seconds. By periodically sending "heartBeat" commands within a period of 30 seconds, you can make sure that no other user applications access the device.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <heartBeat/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <heartBeat/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier

3.3.1.4 setIPConfig

Table 3-9 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V2.2.1	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can change the IP address of the Ident device. If the transferred parameters do not contain any inconsistencies, the device will terminate the connection. This happens as well if the parameters are identical to the previous settings. A reply frame is sent in reply to this command only if there is an error.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <setIPConfig>
       <iPAddress> value_iPAddress </iPAddress>       //opt
       <subNetMask> value_subNetMask </subNetMask>    //opt
       <gateway> value_gateway </gateway>             //opt
       <dNSAddress> value_dNSAddress </dNSAddress>    //*
       <dHCPEnable> value_dHCPEnable </dHCPEnable>    //opt
    </setIPConfig>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <setIPConfig/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_iPAddress	IP address	1.0.0.1... 255.255.255.254	IP address of the device Parameter is ignored if DHCP is activated: value_dHCPEnable = True
value_ subNetMask	IP address	1.0.0.1... 255.255.255.254	Subnet mask of the device Parameter is ignored if DHCP is activated: value_dHCPEnable = True

Parameter	Type	Values	Description
value_gateway	IP address	1.0.0.1... 255.255.255.254	Gateway of the device Parameter is ignored if DHCP is activated: value_dHCPEnable = True
value_ dNSAddress	IP address	1.0.0.1... 255.255.255.254	DNS address of the device Parameter is ignored if DHCP is activated: value_dHCPEnable = True
value_ dHCPEnable	Fixed values	True False	Use of DHCP to assign the IP address.

3.3.1.5 getIPConfig

Table 3-10 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V2.2.1	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can read out the current IP address of the Ident device.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getIPConfig/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getIPConfig>
    <returnValue>
       <iPAddress> value_iPAddress </iPAddress>
       <subNetMask> value_subNetMask </subNetMask>
       <gateway> value_gateway </gateway>
       <dNSAddress> value_dNSAddress </dNSAddress>    //*
```

```
        <dHCPEnable> value_dHCPEnable </dHCPEnable>
      <returnValue/>
      <getIPConfig/>
  </reply>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_iPAddress	IP address	1.0.0.1... 255.255.255.254	IP address of the device Parameter is ignored if DHCP is activated: value_dHCPEnable = True
value_ subNetMask	IP address	1.0.0.1... 255.255.255.254	Subnet mask of the device Parameter is ignored if DHCP is activated: value_dHCPEnable = True
value_gateway	IP address	1.0.0.1... 255.255.255.254	Gateway of the device Parameter is ignored if DHCP is activated: value_dHCPEnable = True
value_ dNSAddress	IP address	1.0.0.1... 255.255.255.254	DNS address of the device Parameter is ignored if DHCP is activated: value_dHCPEnable = True
value_ dHCPEnable	Fixed values	True False	Use of DHCP to assign the IP address.

3.3.2 Device settings

This section describes all the commands with which you can control the configuration of the Ident device (reader, communication module).

3.3.2.1 prepareFwUpdate

Table 3-11 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.1	V4.1	V1.0	V2.1	V1.0	V2.1	V1.0

With this command, you prepare the Ident device for a firmware update and make sure that this is accepted by the device. This command must be the first command within the firmware update process, followed by "transferFW" and "executeFW".

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <prepareFwUpdate>
        <deviceName> value_deviceName </deviceName>     //*
    </prepareFwUpdate>
 </cmd>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <prepareFwUpdate/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ deviceName	Fixed values	Device00 Device01 Device02 Device03 Device04	Name of the device that is connected to the respective interface. "Device00" refers here to the basic device (e.g. RF188C). "Device01"... "Device04" refer to all devices connected to the respective interfaces of the basic device.

3.3.2.2 transferFW

Table 3-12 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.1	V4.1	V1.0	V2.1	V1.0	V2.1	V1.0

With this command, you transfer the firmware data to the Ident device. A maximum of 100 kB data can be transferred with one command. If the data volume of a command is not sufficient, the data is transferred using consecutive commands. Note that the firmware data must be base64-encoded.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <transferFW>
        <fwPartCnt> value_fwPartCnt </fwPartCnt>
        <deviceName> value_deviceName </deviceName>     //*
        <fwData>
            <![CDATA[value_fwData]]>
        </fwData>
    </transferFW>
 </cmd>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <transferFW/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ fwPartCnt	Decimal value 0...9	1...4294967295	Number of consecutive "transferFwUpdate" commands The first specification must start with "1". Consecutive commands must be incremented by "1" in each case.
value_ deviceName	Fixed values	Device00 Device01 Device02 Device03 Device04	Name of the device that is connected to the respective interface. "Device00" refers here to the basic device (communications module, RF600 or RF360R). "Device01"... "Device04" refers to the device connected to the respective interface of the communications module.
value_ fwData	CDATA	--	Firmware data The parameter must be embedded in a CDATA segment base64-encoded. The file size must not exceed 100 kB.

3.3.2.3 executeFW

Table 3-13 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.1	V4.1	V1.0	V2.1	V1.0	V2.1	V1.0

With this command, you activate the transferred firmware on the Ident device.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <executeFW>
        <validateOnly> value_validateOnly </validateOnly>   //opt
        <deviceName> value_deviceName </deviceName>     //*
    </executeFW>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <executeFW/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ validateOnly	Fixed values	True False	Firmware validation/activation True: The firmware is only validated. False: The firmware is validated and activated.
value_ deviceName	Fixed values	Device00 Device01 Device02 Device03 Device04	Name of the device for which the firmware update is performed. "Device00" refers here to the basic device (communications module, RF600 or RF360R). "Device01"... "Device04" refers to the device connected to the respective interface of the communications module.

3.3.2.4 setConfiguration

Table 3-14 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can transfer a configuration to the Ident device. After the configuration is confirmed, it is activated and stored permanently in the flash memory of the Ident device. As an alternative, you can also load a configuration that was created using the WBM.

Note that up to 20 seconds can elapse until a reply frame is received.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <setConfiguration>
        <configData>
        <![CDATA[value_configData]]>
        </configData>
    </setConfiguration>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <setConfiguration>
        <returnValue>
            <configID> value_configID </configID>
        </returnValue>
    </setConfiguration>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ configData	CDATA	--	Configuration data The parameter must be embedded in a CDATA segment.
value_configID	Alphanumeric text	--	Unique identifier of the transferred configuration The ID can also be read out with the "getConfigVersion" command.

3.3.2.5 getConfiguration

Table 3-15 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can read out the configuration saved in the Ident device. You can also use this configuration, for example, to transfer it to other Ident devices.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getConfiguration/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getConfiguration>
        <returnValue>
            <configID> value_configID </configID>
            <configData>
                <![CDATA[value_configData]]>
```

```
            </configData>
        </returnValue>
    </getConfiguration>
  </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ configData	CDATA	--	Configuration data The parameter must be embedded in a CDATA segment.
value_configID	Alphanumeric text	--	Unique identifier of the transferred configuration The ID can also be read out with the "getConfigVersion" command.

3.3.2.6 getConfigVersion

Table 3-16 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can read out the version ID of the saved configuration.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getConfigVersion/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getConfigVersion>
        <returnValue>
            <configID> value_configID </configID>
        </returnValue>
    </getConfigVersion>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_configID	Alphanumeric text	--	Unique identifier of the transferred configuration

3.3.2.7 getActiveConfiguration

Table 3-17 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	--

With this command, you can request the active parameters with which the Ident device is currently working as a configuration file from the Ident device.

Note that the active parameter values may differ from the stored values.

Note that up to 20 seconds can elapse until a reply frame is received.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
        <getActiveConfiguration/>
```

```
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getActiveConfiguration>
        <returnValue>
            <configID> value_configID </configID>
            <configData>
                <![CDATA[value_configData]]>
            </configData>
        </returnValue>
    </getActiveConfiguration>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ configData	CDATA	--	Configuration data The parameter must be embedded in a CDATA segment.
value_configID	Alphanumeric text	--	Unique identifier of the transferred configuration The ID can also be read out with the "getConfigVersion" command.

3.3.2.8 getLogfile

Table 3-18 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can read out the logbook of the Ident device.

Note that up to 20 seconds can elapse until a reply frame is received.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getLogfile>
        <logType> value_logType </logType>
    </getLogfile>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getLogfile>
        <returnValue>
            <logData>
                <![CDATA[value_configData]]>
            </logData>
        </returnValue>
    </getLogfile>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_logType	Fixed values	Diagnosis Logbook Headline ServiceLog Messages	Type of diagnostics file • Diagnosis: File with logbook entries in CSV format including header. Required with older device versions for reasons of compatibility. RF600: The logbook is read out without header and with the "|" character as column separator. • Logbook (default): File with logbook entries in CSV format including header. Recommended with newly created applications. • Headline: Only returns the header line. Recommended for logbook events. • ServiceLog: Service log file with logbook entries in CSV format without header. Siemens technical personnel are responsible for evaluating the log. • Messages: Startup messages of the device. Siemens technical personnel are responsible for evaluating this data
value_logData	CDATA	--	Data logged by the Ident device The parameter must be embedded in a CDATA segment.

Output format of the "Logbook" diagnostics file

The diagnostics file of the type "Logbook" consists of individual lines of text that are divided into multiple columns by a semicolon ";". A header describing the columns contained is always output as the first line. This formatting enables you to open the file in a spreadsheet program, for example, which improves the readability of the data.

Depending on the Ident device used and its configuration (in the WBM), different diagnostic information is output.

	A	B	C	D	E	F	G	
1	Date_Time	Type	Entry	Command_Type	Readpoint	Call_Param	Return	EPCID_UID
2	2019-06-18T08:11:08.849+00:00	COMMANDS	hostGreetings	CMD		version=V2.0,version	version=V3.0,configl	
3	2019-06-18T08:11:08.849+00:00	COMMANDS	hostGreetings_1	CMD		...=V2.1,version=V2.2,v	...D=5D01041D	
4	2019-06-18T08:11:08.850+00:00	COMMANDS	hostGreetings_2	CMD		...ersion=V3.0		

Figure 3-1 Example of a "Logbook" diagnostics file

A logbook entry can only contain up to 6 kB data. When commands are recorded, multiple logbook entries can therefore be created for a single command. The names of the respective commands and a command-specific consecutive number are inserted in the "Entry" column.

The following table provides an overview of the parameters that can be saved in the logbook entries of the diagnostics file depending on the configuration of your diagnostics.

Table 3-19 Parameters contained in every logbook entry

Parameter	Type	Values	Description
Date_Time	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2019-06-18T08:11:08.849+00:00
Type	Fixed values	COMMON ERRORS EVENTS COMMANDS FILTER GPIO	Type of entry The configuration of the Ident device (WBM) determines which events/ message types are entered in the logbook. They are output accordingly in the diagnostics file.
Entry	Text	--	Text of the logbook entry With a logbook entry of the type "EVENTS", "COMMANDS" and "FILTER", the diagnostics file is supplemented by additional logbook entries (see the following table).

Table 3-20 Parameters contained in a logbook entry due to the "EVENTS", "COMMANDS" and "FILTER" parameters

Parameter	Type	Values	Description
Command_Type	Fixed values	CMD CMD_CALL CMD_CHN CMD_RPT CMD_CHNRPT CMD_FAIL RPL RPL_CHN RPL_RPT PRL_CHNRPT RPL_FAIL UNKNOWN	Type of the command Is only filled when the "COMMANDS" message type is activated. With all other message types, this column contains no entries.
Readpoint	Text	--	Name of the read point affected by the entry. If no read point can be assigned (e.g. with the XML command "getReaderStatus"), this column contains no entries.

Parameter	Type	Values	Description
Call_Param	Text	--	Specification of the command parameters Is only filled when the "COMMANDS" message type is activated. With all other message types, this column contains no entries.
Return	Text	--	Specification of the return values Is only filled when the "COMMANDS" message type is activated.
EPCID_UID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format" Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".

Table 3-21 Parameters contained in a logbook entry only with RF600

Parameter	Type	Values	Description
TagPC	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
RSSI	Decimal value 0...9	0...255	RSSI value
Power	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
Antenna	Fixed values	Antenna01 Antenna02 Antenna03 Antenna04	Name of the antenna
Polarization	Fixed values	Circular Linear_vertical Linear_ horizontal Unknown	Polarization of the antenna With external antennas, the parameter is always unknown.
Channel	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
Command_ Retry	Decimal value 0...9	0...65535	Number of command repetitions that were needed before the reader received correct data. "0" corresponds to no attempt. The maximum number of attempts is set using the WBM.

Parameter	Type	Values	Description
Air_Retry	Decimal value 0...9	0...65535	Number of air interface command repetitions that were needed before the reader received correct data. "0" corresponds to no attempt. The maximum number of attempts is specified in the reader firmware and cannot be modified.
FilterData	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.
Inventoried	Decimal value 0...9	0...65535	Indicates how often the transponder was identified via the air interface before it changed to the "Observed" status.
RSSI_Min	Decimal value 0...9	0...255	The lowest RSSI value with which the transponder was detected. This value has no unit and is not directly related to the performance.
RSSI_Max	Decimal value 0...9	0...255	The highest RSSI value with which the transponder was detected. This value has no unit and is not directly related to the performance.

3.3.2.9 resetLogfile

Table 3-22 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can delete all logbook entries.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <resetLogfile>
        <logType> value_logType </logType>    //opt
    </resetLogfile>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <resetLogfile/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_logType	Fixed value	Diagnosis Logbook ServiceLog	Type of diagnostics file • Diagnosis (default with RF640R/ RF670R): Logbook file. Required with older device versions for reasons of compatibility. • Logbook (default): Logbook file. Recommended with newly created applications. • ServiceLog: Service log file. Siemens technical personnel are responsible for evaluating the log.

3.3.2.10 setLogEntry

Table 3-23 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.2	V4.3	V1.2	V2.3	V1.1	V2.3	V1.2

This command allows you to send an entry with log information to the device. Depending on the currently enabled logbook types and log event settings, a logEventReport is created and sent.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <setLogEntry>
        <diagLevel> value_diagLevel </diagLevel>
        <logText> value_logText </logText>
        <onlyReport> value_onlyReport </onlyReport>
    </setLogEntry>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <setLogEntry/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_diagLevel	Fixed values	0x40 0x80	Type of entry The configuration of the Ident device (WBM) determines which events/ message types are entered in the log-book. They are output accordingly in the diagnostics file. 0x40: COMMON 0x80: ERRORS
value_logText	Alphanumeric text	--	This text is passed to the LogEventRe-port.
value_onlyReport	Fixed values	True	A LogEventReport is generated and sent. This parameter must be set to True.

3.3.2.11 setParameter

Table 3-24 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can set a specific read point/antenna parameter for the Ident device.

Please note that changes using this command are stored in volatile memory in the configuration of the WBM. As a result, the Ident device works with the parameter value specified with "setParameter" but does not display this in the WBM.

Note that up to 20 seconds can elapse until a reply frame is received.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <setParameter>
        <name> value_name </name>
        <value> value_value </value>
        <objType> value_objType </objType>     //opt
        <objName> value_objName </objName>     //opt
    </setParameter>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <setParameter/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_name	Alphanumeric text	See following table	Name of the supported parameter
value_value	Alphanumeric text	See following table	Parameter value
value_objType	Alphanumeric text	See following table	Specifies the type of the parameter groups to be addressed.
value_objName	Alphanumeric text	See following table	Name of the specific parameter group

Possible values of the alphanumeric parameters

Table 3-25 Parameter values for the RF1xxC/RF12xC/RF360R products

name	value	objType	objName	Description
Power [1]	2...8	Antenna	Antenna11 Antenna21 Antenna31 Antenna41	Detailed information about these parameters can be found in the Product Information "Input parameters for the RF300 system for programming via communications modules".
scanning_ time	0...255	Device	Device01 Device02 Device03 Device04	
param				
option_1				
distance_limit-ing				
multitag				
field_on_ control				
field_on_ time				
LEDOverride [1]	00000000 ... FFFFFFFF	General	General	Behavior of LEDs in the status bar With this parameter, you can define the behavior of the individual LEDs in the status bar and thus overwrite the actual function of the LEDs. Distribution of 32 bits in 3 areas: Bit 31 ... Bit 16: Selection of LED Bit 15 ... Bit 8: Flash frequency of LED Bit 7 ... Bit 0: Color of LED

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment

Table 3-26 Parameter values for the RF600 readers

name	value	objType	objName	Description
Power	0, 5.00...33.00	Antenna	Antenna01 Antenna02 Antenna03 Antenna04	Radiated power of the antenna in [dB] Increment: 0.25 dB
RssiThreshold	0...255	Antenna	Antenna01 Antenna02 Antenna03 Antenna04	RSSI threshold value Transponders with lower RSSI values are not taken into account. This value has no unit and is not directly related to the performance.
Polarization	Default Circular Linear_vertical Linear_ horizontal All	Antenna	Antenna01 Antenna02 Antenna03 Antenna04	Polarization of the antenna This parameter is only relevant with switchable antennas. The "Default" value sets the antenna to linear-vertical.
RssiDelta	0...255	Source	See description	Difference for RSSI values Maximum difference compared with the RSSI value of the transponder with the highest RSSI value that the transponders can have and still be processed. This value has no unit and is not directly related to the performance. "objName" must have the name of the corresponding read point.
Modulation-Scheme	32, 33, 34, 35, 37, 65	General	General	Modulation scheme of the read point This parameter also specifies which transponder types are identified (ISO 18000-62/-63).
LEDOverride	00000000 ... FFFFFFFF	General	General	Behavior of LEDs in the status bar With this parameter, you can define the behavior of the individual LEDs in the status bar and thus overwrite the actual function of the LEDs. Distribution of 32 bits in 3 areas: Bit 31 ... Bit 16: Selection of LED Bit 15 ... Bit 8: Flash frequency of LED Bit 7 ... Bit 0: Color of LED

Table 3-27 Parameter values for the RF69xR readers

name	value	objType	objName	Description
Power	0, 5.00...33.00	Antenna	Antenna01 ... Antenna08	Radiated power of the antenna in [dB] Increment: 0.25 dB
RssiThreshold	-10000...+10000	Antenna	Antenna01 ... Antenna08	RSSI threshold value Transponders with lower RSSI values are not taken into account. RSSI values in 1/100 dBm
Polarization	Default Circular Linear_vertical Linear_horizontal All	Antenna	Antenna01 ... Antenna08	Polarization of the antenna This parameter is only relevant with switchable antennas. The "Default" value sets the antenna to linear-vertical.
RssiDelta	0...20000	Source	See description	Difference for RSSI values Maximum difference compared with the RSSI value of the transponder with the highest RSSI value that the transponders can have and still be processed. RSSI values in 1/100 dBm "objName" must have the name of the corresponding read point.

Table 3-28 Parameter values for the RF1100 readers

name	value	objType	objName	Description
LEDOverride	00000000 ... FFFFFFFF	General	General	Behavior of LEDs in the status bar With this parameter, you can define the behavior of the individual LEDs in the status bar and thus overwrite the actual function of the LEDs. Distribution of 32 bits in 3 areas: Bit 31 ... Bit 16: Selection of LED Bit 15 ... Bit 8: Flash frequency of LED Bit 7 ... Bit 0: Color of LED
Bluetooth	True False	Device	Device01	Switch on / switch off Bluetooth functionality
CardFamilies-Filter	0000 ... FFFF	Device	Device01	Selection of which card types are addressed. Hexadecimal 16-bit value FFFF (default): All card types F000: RFU 0800: LEGIC Prime 0400: BluetoothMce 0200: 125 kHz Part 2 0100: SRIX 0080: 125 kHz Part 1 0040: Felica 0020: IClass 0010: IClass ISO14B 0008: ISO14443B 0004: ISO15693 0002: RFU 0001: ISO1443A

Table 3-29 LEDOverride: Behavior of the LEDs

Bits	Description	
Bit 31 ... Bit 16	Selection of LED	
	RF360R • Bit 29: Reader LEDs	RF68xR (status display) • Bit 16: LED 1 (right) • ... • Bit 24: LED 9 (left) RF61xR • Bit 25: "PRE" LED
	RF1100R • Bit 16: LED • Bit 17: Acoustic signal	

Bits	Description
Bit 15 ... Bit 8	Flash frequency of LED • 0: Static • 1: 1 Hz flash frequency • 2: 2 Hz flash frequency • 3: Reserved • 4: 4 Hz flash frequency • 5 ... 255: Reserved
Bit 7 ... Bit 0	Color of LED • 0: Transparent (the reader controls the LED) • 1: Off • 2: Red • 3: Green • 4: Yellow • 5: Magenta [1)] • 6: Cyan [1)] • 7: White [1)] • 8 ... 255: Reserved

[1)] Only with RF360R readers

3.3.2.12 getParameter

Table 3-30 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can set a specific read point/antenna parameter of the Ident device. The return value contains the value currently being used.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getParameter>
        <name> value_name </name>
        <objType> value_objType </objType>
        <objName> value_objName </objName>
    </getParameter>
 </cmd>
```

```
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getParameter>
        <returnValue>
            <value> value_value </value>
        </returnValue>
    </getParameter>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_name	Alphanumeric text	See following table	Name of the supported parameter
value_value	Alphanumeric text	See following table	Parameter value
value_objType	Alphanumeric text	See following table	Specifies the type of the parameter groups to be addressed.
value_objName	Alphanumeric text	See following table	Name of the specific parameter group

Possible values of the alphanumeric parameters

Table 3-31 Parameter values for the RF1xxC/RF12xC/RF360R products

name	value	objType	objName	Description
Power [1]	2...8	Antenna	Antenna11 Antenna21 Antenna31 Antenna41	Detailed information about these parameters can be found in the Product Information "Input parameters for the RF300 system for programming via communications modules".
scanning_ time	0...255	Device	Device01 Device02 Device03 Device04	
param				
option_1				
distance_limit-ing				
multitag				
field_on_ control				
field_on_ time				
LEDOverride [1]	00000000 ... FFFFFFFF	General	General	Behavior of LEDs in the status bar With this parameter, you can define the behavior of the individual LEDs in the status bar and thus overwrite the actual function of the LEDs. Distribution of 32 bits in 3 areas: Bit 31 ... Bit 16: Selection of LED Bit 15 ... Bit 8: Flash frequency of LED Bit 7 ... Bit 0: Color of LED

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment

Table 3-32 Parameter values for the RF600 readers

name	value	objType	objName	Description
Power	0, 5.00...33.00	Antenna	Antenna01 Antenna02 Antenna03 Antenna04	Radiated power of the antenna in [dB] Increment: 0.25 dB
RssiThreshold	0...255	Antenna	Antenna01 Antenna02 Antenna03 Antenna04	RSSI threshold value Transponders with lower RSSI values are not taken into account. This value has no unit and is not directly related to the performance.

name	value	objType	objName	Description
Polarization	Default Circular Linear_vertical Linear_horizontal All	Antenna	Antenna01 Antenna02 Antenna03 Antenna04	Polarization of the antenna This parameter is only relevant with switchable antennas. The "Default" value sets the antenna to linear-vertical.
RssiDelta	0...255	Source	See description	Difference for RSSI values Maximum difference compared with the RSSI value of the transponder with the highest RSSI value that the transponders can have and still be processed. This value has no unit and is not directly related to the performance. "objName" must have the name of the corresponding read point.
Modulation-Scheme	32, 33, 34, 35, 37, 65	General	General	Modulation scheme of the read point This parameter also specifies which transponder types are identified (ISO 18000-62/-63).
LEDOverride	00000000 ... FFFFFFFF	General	General	Behavior of LEDs in the status bar With this parameter, you can define the behavior of the individual LEDs in the status bar and thus overwrite the actual function of the LEDs. Distribution of 32 bits in 3 areas: Bit 31 ... Bit 16: Selection of LED Bit 15 ... Bit 8: Flash frequency of LED Bit 7 ... Bit 0: Color of LED

Table 3-33 Parameter values for the RF69xR readers

name	value	objType	objName	Description
Power	0, 5.00...33.00	Antenna	Antenna01 ... Antenna08	Radiated power of the antenna in [dB] Increment: 0.25 dB
RssiThreshold	-10000...+10000	Antenna	Antenna01 ... Antenna08	RSSI threshold value Transponders with lower RSSI values are not taken into account. RSSI values in 1/100 dBm

name	value	objType	objName	Description
Polarization	Default Circular Linear_vertical Linear_ horizontal All	Antenna	Antenna01 ... Antenna08	Polarization of the antenna This parameter is only relevant with switchable antennas. The "Default" value sets the antenna to linear-vertical.
RssiDelta	0...20000	Source	See description	Difference for RSSI values Maximum difference compared with the RSSI value of the transponder with the highest RSSI value that the transponders can have and still be processed. RSSI values in 1/100 dBm "objName" must have the name of the corresponding read point.

Table 3-34 Parameter values for the RF1100 readers

name	value	objType	objName	Description
LEDOverride	00000000 ... FFFFFFFF	General	General	Behavior of LEDs in the status bar With this parameter, you can define the behavior of the individual LEDs in the status bar and thus overwrite the actual function of the LEDs. Distribution of 32 bits in 3 areas: Bit 31 ... Bit 16: Selection of LED Bit 15 ... Bit 8: Flash frequency of LED Bit 7 ... Bit 0: Color of LED
Bluetooth	True False	Device	Device01	Switch on / switch off Bluetooth functionality

name	value	objType	objName	Description
Licenses	00000000 FFFFFFFF	Device	Device01	32-bit as hexadecimal value Each bit corresponds to one license: Bit 0 = Hid
CardFamilies-Filter	0000 ... FFFF	Device	Device01	Selection of which card types are addressed. Hexadecimal 16-bit value FFFF (default): All card types F000: RFU 0800: LEGIC Prime 0400: BluetoothMce 0200: 125 kHz Part 2 0100: SRIX 0080: 125 kHz Part 1 0040: Felica 0020: IClass 0010: IClass ISO14B 0008: ISO14443B 0004: ISO15693 0002: RFU 0001: ISO1443A

3.3.2.13 setParameterList

Table 3-35 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	V4.0	V1.0	V2.0	V1.0	V2.0	V1.0

With this command, you can set multiple read points/antenna parameters for the Ident device. If an error occurs with a parameter of the list, this has the result that no single parameter of the list is set.

Please note that changes using this command are stored in volatile memory in the configuration of the WBM. As a result, the Ident device works with the parameter value specified with "setParameterList" but does not display this in the WBM.

Note that up to 20 seconds can elapse until a reply frame is received.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <setParameterList>
      <param>
        <name> value_name </name>
```

```
        <value> value_value </value>
        <objType> value_objType </objType>     //opt
        <objName> value_objName </objName>     //opt
      </param>
        ...
      <param>
        ...
      </param>
    </setParameterList>
  </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <setParameterList/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

You can find a detailed description of the parameters in the section "setParameter (Page 45)".

3.3.2.14 getParameterList

Table 3-36 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	V4.0	V1.0	V2.0	V1.0	V2.0	V1.0

With this command, you can set multiple read points/antenna parameters of the Ident device. The return value contains the value currently being used.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getParameterList>
      <param>
        <name> value_name </name>
        <objType> value_objType </objType>
        <objName> value_objName </objName>
      </param>
        ...
      <param>
        ...
      </param>
    </getParameterList>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getParameterList>
        <returnValue>
          <param>
            <name> value_name </name>
            <value> value_value </value>
            <objType> value_objType </objType>
            <objName> value_objName </objName>
          </param>
            ...
          <param>
            ...
          </param>
        </returnValue>
    </getParameterList>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

You can find a detailed description of the parameters in the section "getParameter (Page 50)".

3.3.2.15 setIMParameter

Table 3-37 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.1	--	V1.0	--	V1.0	--	V1.0

With this command, you can write a specific I&M parameter to the Ident device once.

NOTICE
Do not enable use of this command This function is not released for use in a user application. Only use this function under instruction from a Siemens employee.

With RF69xR and RF1100, note that the data can only be written once. Every additional attempt to overwrite the data is rejected with an error. The validity of a newly written value can be checked with "getIMParameter" after restart of the module. A module restart is required to ensure that the value is recognized or processed by the system.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <setIMParameter>
        <name> value_name </name>
        <value> value_value </value>
    </setIMParameter>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <setIMParameter/>
```

```
  </reply>
 </frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_name	Alphanumeric text	See following table	Name of the supported parameter
value_value	Alphanumeric text	See following table	Parameter value

Possible values of the alphanumeric parameters

Note

Checking validity of written values

The validity of a value written via "setIMParameter" can only be checked with "getIMParameter" after restart of the module.

Table 3-38 Parameter values of the alphanumeric parameters

name	value	Description
MacAddress0	Hexadecimal value 0...9, A...F	MAC address of the basic device 16 hex characters without separator e.g.: 11AA22BB33CC
MacAddress1	Hexadecimal value 0...9, A...F	MAC address of the first network port 16 hex characters without separator e.g.: 11AA22BB33CC
MacAddress2	Hexadecimal value 0...9, A...F	MAC address of the second network port 16 hex characters without separator e.g.: 11AA22BB33CC
SerialNumber	Text	Device serial number 10 text characters e.g.: VFE1821961
HwVersion [1)]	Decimal value 0...9	Hardware version of the device
Mlfb [1)]	Text	MLFB of the basic device with blank line after the four characters For example: 6GT2 811-7AD20-0AA0

[1)] Depending on the Ident device used, the FW version or the WBM parameter assignment

3.3.2.16 getIMParameter

Table 3-39 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.1	--	V1.0	--	V1.0	--	V1.0

With this command, you can set a specific I&M parameter of the Ident device. The return value contains the value currently being used.

NOTICE
Do not enable use of this command This function is not released for use in a user application. Only use this function under instruction from a Siemens employee. Use the "getDeviceStatus()" function to query parameters.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getIMParameter>
        <name> value_name </name>
    </getIMParameter>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getIMParameter>
        <returnValue>
            <value> value_value </value>
        </returnValue>
    </getIMParameter>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_name	Alphanumeric text	See following table	Name of the supported parameter
value_value	Alphanumeric text	See following table	Parameter value

Possible values of the alphanumeric parameters

Note

Checking validity of written values

The validity of a value written via "setIMParameter" can only be checked with "getIMParameter" after restart of the module.

Table 3-40 Parameter values of the alphanumeric parameters

name	value	Description
MacAddress0	Hexadecimal value 0...9, A...F	MAC address of the basic device 16 hex characters without separator e.g.: 11AA22BB33CC
MacAddress1	Hexadecimal value 0...9, A...F	MAC address of the first network port 16 hex characters without separator e.g.: 11AA22BB33CC
MacAddress2	Hexadecimal value 0...9, A...F	MAC address of the second network port 16 hex characters without separator e.g.: 11AA22BB33CC
SerialNumber	Text	Device serial number 10 text characters e.g.: VFE1821961
HwVersion [1)]	Decimal value 0...9	Hardware version of the device
Mlfb [1)]	Text	MLFB of the basic device with blank line after the four characters For example: 6GT2 811-7AD20-0AA0

[1)] Depending on the Ident device used, the FW version or the WBM parameter assignment

3.3.2.17 setTime

Table 3-41 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can set the internal clock of the Ident device.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <setTime>
        <utcTime> value_utcTime </utcTime>
    </setTime>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <setTime/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00. Note: The reader only accepts time information from 01.01.2000 ... 18.01.2038.

3.3.2.18 getTime

Table 3-42 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can read out the current time stamp of the internal device clock time.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getTime/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getTime>
        <returnValue>
            <utcTime> value_utcTime </utcTime>
        </returnValue>
    </getTime>
```

```
  </reply>
 </frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.

3.3.2.19 setIO

Table 3-43 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	--	--	--

With this command, you can set the digital outputs of the Ident device.

The general settings for the behavior of the digital outputs such as "Inactivity" or "Reset time" are specified in a configuration using the WBM. You can find detailed information on the configuration of the digital outputs in the WBM sub-section "Digital outputs" of the relevant Ident device manual.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <setIO>
        <outValue> value_outValue </outValue>
    </setIO>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <setIO/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_outValue	Characters 0, 1, X	00000000.... XXXXxxxx... 11111111	Each position stands for a digital output of the Ident device: • Output0: 1st position (least significant bit right) • Output1: 2nd position • ... • Output7: 8th position Depending on the value of the particular position, the corresponding output is set to ON (1) or OFF (0) or remains unchanged (X). Example: A "value_outValue" of "0X11" • Sets Output0 to ON • Sets Output1 to ON • Leaves Output2 unchanged • Sets Output3 to OFF This command outputs a negative reply "ERROR_PARAMETER_OUT_OF_RANGE" if there are more outputs to be set than the Ident device supports. If, for example, the Ident device only supports 2 digital outputs, no output will be set in the example above in which 4 outputs are addressed.

3.3.2.20 getIO

Table 3-44 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	--	--	--

With this command, you can read out the current states of all digital inputs/outputs of the Ident device.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getIO/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getIO>
        <returnValue>
            <inValue> value_inValue </inValue>
            <outValue> value_outValue </outValue>
        </returnValue>
    </getIO>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_inValue	Binary value 0, 1	00000000.... 11111111	Each position stands for a digital input of the Ident device: • Input0: 1st position (least significant bit right) • Input1: 2nd position • ... • Input7: 8th position Depending on the value of the particular position, the corresponding input is set to ON (1) or OFF (0). If the Ident device does not support digital inputs/outputs, no value is output.
value_outValue	Binary characters 0, 1	00000000.... 11111111	Each position stands for an output of the Ident device: • Output0: 1st position (least significant bit right) • Output1: 2nd position • ... • Output7: 8th position Depending on the value of the particular position, the corresponding output is set to ON (1) or OFF (0). If the Ident device does not support digital inputs/outputs, no value is output.

3.3.2.21 resetReader

Table 3-45 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can reset the reader to the factory settings or restart the reader.

After a positive reply from the user application, the reader terminates the TCP/IP connection and then performs a reset. Note that, following this, you need to establish the connection again via the user application and restart the reader with the "hostGreetings" command.

From XML API protocol version V3.0, this command is still present for reasons of compatibility. This command has been replaced by the "resetDevice" command.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <resetReader>
        <resetType> value_resetType </resetType>      //opt
    </resetReader>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

Note that the reply frame may not be sent if the reader restarts.

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <resetReader/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
resetType	Fixed values	Reset2Factory Reboot Defcfg SwitchUHFTo422 RebootUHF	Type of reset Reset2Factory: Reset to the factory settings and delete the stored configuration. Reboot (default): Hardware reset of the reader without deleting the stored configuration. After the restart, communication must be re-established. Defcfg: Reset to default configuration. Settings on IP address, user management and log files are retained. The reader is not restarted. Additionally for RF600: SwitchUHFTo422 and RebootUHF: Only present for certification reasons. These values should not be used in regular operation.

3.3.2.22 resetDevice

Table 3-46 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can reset the device to the factory settings.

After a positive reply from the user application, the device terminates the TCP/IP connection and then performs a reset. Note that, following this, you need to establish the connection again via the user application and restart the device with the "hostGreetings" command.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <resetDevice>
        <resetType> value_resetType </resetType>      //opt
    </resetDevice>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <resetDevice/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
resetType	Fixed values	Reset2Factory Reboot Defcfg SwitchUHFTo422 RebootUHF	Type of reset Reset2Factory: Reset to the factory settings and delete the stored configuration. Reboot (default): Hardware reset of the device without deleting the stored configuration. After the restart, communication must be re-established. Defcfg: Reset to default configuration. Settings on IP address, user management and log files are retained. The reader is not restarted. Additionally for RF600: SwitchUHFTo422 and RebootUHF: Only present for certification reasons. These values should not be used in regular operation.

3.3.2.23 getReaderStatus

Table 3-47 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can read out status information from the reader.

From XML API protocol version V3.0, this command is still present for reasons of compatibility. This command has been replaced by the "getDeviceStatus" command.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getReaderStatus/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getReaderStatus>
        <returnValue>
            <readerType> value_readerType </readerType>
            <mLFB> value_mLFB </mLFB>
            <hWVersion> value_hWVersion </hWVersion>
            <fWVersion> value_fWVersion </fWVersion>
            <serialNumber> value_serialNumber </serialNumber>
            <subVersions>                                //*
                <version> value_version </version>
                ...
            </subVersions>
        </returnValue>
    </getReaderStatus>
 </reply>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ readerType	Fixed values	SIMATIC_RF610R SIMATIC_RF615R SIMATIC_RF650R SIMATIC_RF680R SIMATIC_RF685R SIMATIC_RF690R SIMATIC_RF695R SIMATIC_RF166C SIMATIC_RF185C SIMATIC_RF186C SIMATIC_RF188C SIMATIC_RF186CI SIMATIC_RF188CI SIMATIC_RF128C SIMATIC_RF360R SIMATIC_RF1140R SIMATIC_RF1170R	Device type
value_mLFB	Alphanumeric text	--	Siemens article number of the device Example: 6GT2811-6AA10-0AA0
value_hWVersion	Alphanumeric text	--	Hardware version of the device
value_fWVersion	Alphanumeric text	--	Firmware version of the device
value_ serialNumber	Alphanumeric text	--	Serial number of the device
value_version	Alphanumeric text	--	Reader-specific version of components Note that the number of sub-versions can change. In future versions, there may be more or less sub-versions.

3.3.2.24 getDeviceStatus

Table 3-48 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can read out status information from the device.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getDeviceStatus/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getDeviceStatus>
        <returnValue>
            <readerType> value_readerType </readerType>
            <mLFB> value_mLFB </mLFB>
            <hWVersion> value_hWVersion </hWVersion>
            <fWVersion> value_fWVersion </fWVersion>
            <serialNumber> value_serialNumber </serialNumber>
            <subVersions>                                //*
                <version> value_version </version>
                ...
            </subVersion>
        </returnValue>
    </getDeviceStatus>
 </reply>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ readerType	Fixed values	SIMATIC_RF610R SIMATIC_RF615R SIMATIC_RF650R SIMATIC_RF680R SIMATIC_RF685R SIMATIC_RF690R SIMATIC_RF695R SIMATIC_RF166C SIMATIC_RF185C SIMATIC_RF186C SIMATIC_RF188C SIMATIC_RF186CI SIMATIC_RF188CI SIMATIC_RF128C SIMATIC_RF360R SIMATIC_RF1140R SIMATIC_RF1170R	Device type
value_mLFB	Alphanumeric text	--	Siemens article number of the device Example: 6GT2811-6AA10-0AA0
value_hWVersion	Alphanumeric text	--	Hardware version of the device
value_fWVersion	Alphanumeric text	--	Firmware version of the device
value_serialNum-ber	Alphanumeric text	--	Serial number of the device
value_version	Alphanumeric text	--	Reader-specific version of components Note that the number of sub-versions can change. In future versions, there may be more or less sub-versions.

3.3.2.25 getCMReaderStatus

Table 3-49 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	--	--	V1.3	V1.0	V1.0	--

With this command, you can read out status information from the CM/reader. This includes, for example, various statistics counters of the devices currently connected to the communications module. This information can be requested either from a specific device or from all devices.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getCMReaderStatus>
       <deviceName> value_deviceName </deviceName>   //opt
       <mode> value_mode </mode>                     //opt
    </getCMReaderStatus>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getCMReaderStatus>
        <returnValue>
            <cmReaderStatus>
               <cmReaderStatusData> value_cmReaderStatusData </
cmReaderStatusData>
               <deviceName> value_deviceName </deviceName>
               <values>                        //*
                 <value> value_value </value>
                  ...
               </values>
            </cmReaderStatus>
            ...
        </returnValue>
    </getCMReaderStatus>
 </reply>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ deviceName	Fixed values	Device01 Device02 Device03 Device04	Name of the device that is connected to the respective interface. If this parameter is not set, all interfaces/devices are selected automatically.
value_mode	Fixed values	1 6 235...255	Mode of the status data to be requested. The supported values depend on the reader type assigned to the specified read point. If this parameter is not set, the value "1" is used automatically. The meaning of the values depends on the connected device and the block used. You can find additional information in the relevant block manual.
value_ cmReaderStatus Data	Hexadecimal value 0...9, A...F	--	The raw reply data of the "getCMReader" command as simple byte sequence. It is mode-specific and reader-specific.
value_value	Alphanumeric text	--	Named specific values from the raw reply data. Note that the number of values can change. There may be more or fewer values in future versions.

3.3.2.26 getAllSources

Table 3-50 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can query the names of all configured read points of the device. You can change the names of the read points in the WBM of the respective device.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getAllSources/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getAllSources>
        <returnValue>
            <sourceName> value_sourceName </sourceName>
            ...
            <sourceName> value_sourceName </sourceName>    //*
        </returnValue>
    </getAllSources>
 </reply>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").

3.3.2.27 getCLPConfig

Table 3-51 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.2	--	V1.0	--	--	--	--

With this command, you can read out information about the inserted CLP.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
```

```
    <getCLPConfig/>
  </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getCLPConfig>
        <returnValue>
            <clpStatus> value_clpStatus </clpStatus>
            <clpDevGroup> value_clpStatus </clpDevGroup>
            <clpDevType> value_clpStatus </clpDevType>
            <clpFwVersion> value_clpStatus </clpFwVersion>
            <clpFwIncluded> value_clpStatus </clpFwIncluded>
            <clpInfo> value_clpInfo </clpInfo>
        </returnValue>
    </getCLPConfig>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ clpStatus	Fixed values	ACCEPTED NOT ACCEPTED NOT PRESENT FACTORY DISABLED	Status of the CLP ACCEPTED: CLP is configured as a back-up medium NOT ACCEPTED: Invalid or incompatible configuration of the CLP NOT PRESENT: No CLP inserted FACTORY: CLP when delivered DISABLED: CLP functionality deactivated
value_ clpDevGroup	Fixed values	SIMATIC IDENT RF600	Device family to which the data stored on the CLP corresponds.

Parameter	Type	Values	Description
value_ clpDevType	Fixed values	SIMATIC RF690R ETSI SIMATIC RF690R FCC SIMATIC RF690R CMIIT SIMATIC RF695R ETSI SIMATIC RF695R FCC SIMATIC RF695R CMIIT	Device type to which the data stored on the CLP corresponds.
value_ clpFwVersion	Text	--	Firmware version which is stored on the CLP.
value_ clpFwIncluded	Text	True False	Indication of whether firmware is stored on the CLP.
value_ clpInfo	Text	--	Additional information on the device type and CLP

3.3.3 Transponder processing

This section describes all the commands with which you can control the processing of the transponder data. There are two ways of querying transponder data:

- Synchronous transponder command
  Commands that return the transponder data in the reply frame.
  The reader executes the required action once and then sends back the acquired transponder data.
  Algorithms that work with individual transponder commands such as "Read/WritePowerBoost" and "Read/WriteRetry" are active.
- Asynchronous transponder events
  "TagEventReports" are sent by the reader to the user application on the reader's initiative. The transponder data is acquired only by triggers of the read point. The configuration of the reader opens up numerous options for the trigger configuration of a read point.

You specify the message content contained in the reply frames using the "Tag events" in the WBM. You will find more information on the tag events in the WBM sub-section "Communication" of the relevant Ident device manual.

Note

Transponder processing with connection via RF1xxC and RF360R

If you want to process transponders via RF1xxC and RF360R, you need to activate the presence check beforehand.

3.3.3.1 editBlackList

Table 3-52 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V2.0	V1.0	V1.0	--	--	--	--

With this command, you save EPC IDs to/remove them from the "Black list".

The black list is a filter mechanism with which transponders are filtered out. Transponders whose IDs are stored in the black list are ignored and not processed. Please note that the black list is a circular buffer with a configurable size. If all the entries in the black list are occupied, the next new entry deletes the oldest entry.

You specify the size of the black list buffer in the WBM. You will find more information on the black list in the WBM sub-section "Read points" of the relevant Ident device manual.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <editBlacklist>
        <sourceName> value_sourceName </sourceName>
        <blackListCmd> value_blackListCmd </blackListCmd>
        <tagID> value_tagID </tagID>              //opt
           ...
        <tagID> value_tagID </tagID>              //opt
    </editBlacklist>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <editBlacklist/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_ black-ListCmd	Fixed values	Add Add_obs Del Del_all	How "setBlacklist" works: • Add: All the following EPC IDs will be saved • Add_obs: All EPC IDs with the "observed" status will be saved in the black list. • Del: All the following EPC IDs will be removed from the black list • Del_all: All entries in the black list will be removed.
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format" Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".

3.3.3.2 getBlackList

Table 3-53 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V2.0	V1.0	V1.0	--	--	--	--

With this command, you can read out all EPC IDs currently saved in the "Black list".

The black list is a filter mechanism with which transponders are filtered out. Transponders whose IDs are stored in the black list are ignored and not processed.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
```

```
    <getBlacklist>
        <sourceName> value_sourceName </sourceName>
    </getBlacklist>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getBlacklist>
        <returnValue>
           <tagID> value_tagID </tagID>          //*
           ...
           <tagID> value_tagID </tagID>          //*
        </returnValue>
    </getBlacklist>
 </reply>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format" Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".

3.3.3.3 triggerSource

Table 3-54 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

Please note that this command was designed for transponder processing using the RF600 readers. This command is also present with RF1xxC and RF360R for compatibility reasons, but application of it is generally not useful, except when you want to read out the UIDs of the transponders.

With this command, you trigger inventories on the selected read points. In this case, detected transponders are subjected to the smoothing algorithm and can have the different statuses "glimpsed", "observed" and "lost". The data of the identified statuses is sent to the XML user application as "TagEventReport".

The configuration parameters of the read point such as "Smoothing" and the definition of the data of each transponder (tag fields, RSSI value, ...) to be sent are taken from the stored configuration. You set this parameter in the WBM. You will find more information on the parameters in the WBM sub-section "Read points" of the relevant Ident device manual (RF600).

Note that there is no smoothing algorithm with RF1xxC. When a transponder is detected with RF1xxC/RF360R, an "observed" event is signaled, every time a transponder leaves the antenna field a "lost" event is generated.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <triggerSource>
        <sourceName> value_sourceName </sourceName>
        <triggerMode> value_triggerMode </triggerMode>   //opt
    </triggerSource>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
```

```
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <triggerSource/>
  </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_ triggerMode	Fixed values	Single Start Stop	Trigger type With RF600, the duration and quantity can be determined via the WBM. Single (default): Single triggering of the read point. Start: The read point is continuously triggered until a stop command is sent. Stop: Stops triggering of the read point. Note that this command is only effective for previously executed trigger commands. If a continuous trigger was configured with RF600, this command does not affect it.

3.3.3.4 readTagIDs

Table 3-55 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you trigger inventories on the selected read points. A reply frame returns information on which transponders were detected. If no transponder was detected, a positive reply is also returned but it does not contain transponder data. With RF600, multiple transponders can be read and their IDs can be output. With all other devices, only an ID is output, even if there are multiple transponders in the antenna field.

Note that this command cannot be interrupted with RF600. The number of inventories performed and the time period over which the reader performs inventories depend on the "value_duration" and "value_unit" parameters. An acknowledge frame from the reader is only sent after this. You must consider these reply times in the user application. The configuration

parameters of the read point (read cycles per trigger, read timeout, ...) are not used. You will find more information on the parameters in the WBM sub-section "Read points" of the relevant Ident device manual.

Note

RF600: Filter mechanisms influence the results

Please note that filter mechanisms defined with RF600 affect the results. With an active filter, only the transponders that have not been filtered out are listed in the reply frame.

You will find more information on the filter mechanisms in the WBM sub-section "Filters" of the relevant Ident device manual.

Note

Delay of the reply frames

Please note that the reply frame of the Ident device may be delayed further due to the configured duration of the command.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <readTagIDs>
        <sourceName> value_sourceName </sourceName>
        <duration> value_duration </duration>      //opt
        <unit> value_unit </unit>                  //opt
    </readTagIDs>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id>value_id</id>
    <resultCode> 0 </resultCode>
    <readTagIDs>
        <returnValue>
            <tag>
                <tagID> value_tagID </tagID>
                <sourceName> value_sourceName </sourceName>
```

```
                <tagPC> value_tagPC </tagPC>            //*
                <tagXPC> value_tagXPC </tagXPC>         //*
                <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                <phase> value_phase </phase>            //*
                <phaseEnd> value_phaseEnd </phaseEnd>   //*
                <utcTime> value_utcTime </utcTime>      //opt
                <antennaName> value_antennaName </antennaName>     //*
                <rSSI> value_rSSI </rSSI>               //*
                <channel> value_channel </channel>      //*
                <power> value_power </power>            //*
                <polarization> value_polarization </polarization>  //*
                <inventoried> value_inventoried </inventoried>     //*
                <filterDataAvailable> value_filterDataAvailable
                </filterDataAvailable>                  //*
            </tag>
            ...
            <tag>        //opt
               ...
            </tag>       //opt
        </returnValue>
    </readTagIDs>
  </reply>
 </frame>
```

//opt ≙ Optional: Parameter can be omitted. In the reply, the parameter is transferred depending on the WBM parameter assignment ("Settings - Communication").

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_duration	Decimal value 0...9	0...65535	Duration for which the selected read point will read transponders. If the value "0" is set or the parameter is empty, only one read cycle is executed.
value_unit	Fixed values	Time Count	Specifies the unit for the duration. Time (default): Time in milliseconds Count: Number of inventories

Parameter	Type	Values	Description
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format" Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC [1]	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
value_channel [1]	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power [1]	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB

Parameter	Type	Values	Description
value_ polarization [1]	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. The value "Default" ≙ "Linear_vertical".
value_inventoried [1]	Decimal value 0...9	0...65535	Indicates how often the transponder was identified via the air interface in this command.
value_ filterDataAvailable [1]	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment
[2] Only for the RF61xR/RF650R/RF68xR readers
[3] Only for the RF69xR readers

3.3.3.5 writeTagID

Table 3-56 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	--	--	--	--

With this command, you write a new ID to a transponder. To ensure clear identification when writing the ID, either there can be only one transponder in the antenna field or the previous ID of the transponder needs to be specified. If there is more than one transponder in the antenna field or if no transponder ID is specified, a negative response is returned.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <writeTagID>
        <sourceName> value_sourceName </sourceName>
        <tagID> value_tagID </tagID>            //opt
        <newID> value_newID </newID>
        <idLength> value_idLength </idLength>   //opt
        <password> value_password </password>   //opt
    </writeTagID>
 </cmd>
```

```
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <writeTagID>
        <returnValue>
            <tag>
                <tagID> value_tagID </tagID>
                <tagPC> value_tagPC </tagPC>            //opt
                <tagXPC> value_tagXPC </tagXPC>         //opt
                <tagXPC2> value_tagXPC2 </tagXPC2>      //opt
                <phase> value_phase </phase>            //opt
                <phaseEnd> value_phaseEnd </phaseEnd>   //opt
                <utcTime> value_utcTime </utcTime>      //opt
                <antennaName> value_antennaName </antennaName>     //opt
                <rSSI> value_rSSI </rSSI>               //opt
                <channel> value_channel </channel>      //opt
                <power> value_power </power>            //opt
                <polarization> value_polarization </polarization>  //opt
                <commandRetry> value_commandRetry </commandRetry>  //opt
                <airRetry> value_airRetry </airRetry>   //opt
                <filterDataAvailable> value_filterDataAvailable
                </filterDataAvailable>                  //opt
            </tag>
        </returnValue>
    </writeTagIDs>
 </reply>
</frame>
```

//opt ≙ Optional: Parameter can be omitted. In the reply, the parameter is transferred depending on the WBM parameter assignment ("Settings - Communication").

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format". This function accesses all transponders with this ID. If this parameter is empty or not even transferred, the function applies to all transponders. Nevertheless only one transponder is then permitted in the antenna field. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_newID	Hexadecimal value 0...9, A...F	--	New EPC-ID in "RAW Hex Data Format" to be written to the transponder. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_idLength	Decimal value 0...9	16, 32, 48...496	Length of the new EPC-ID in bits. If this parameter is set, the reader checks the correct length of "value_newID". If the check fails, a negative reply is sent. Without this parameter, the new EPC-ID is only checked if the length is a multiple of 16 bits.
value_password	Hexadecimal value 0...9, A...F	00000000.... FFFFFFFF	Access password for the transponder This parameter does not need to be defined if there is no password protection activated for the transponder.
value_tagPC	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".

Parameter	Type	Values	Description
value_tagXPC2	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI	Decimal value 0...9	0...255 [1)] -10000...+10000 [2)]	RSSI value This value has no unit and is not directly related to the performance. [1)] RSSI values in 1/100 dBm [2)]
value_channel	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. The value "Default" ≙ "Linear_vertical".
value_ commandRetry	Decimal value 0...9	0...65535	Number of times the reader repeated commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of attempts is set using the WBM.

Parameter	Type	Values	Description
value_airRetry	Decimal value 0...9	0...65535	Number of times the reader repeated air interface commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of air interface commands is specified in the reader firmware and cannot be modified.
value_ filterDataAvailable	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.

1) Only for the RF61xR/RF650R/RF68xR readers

2) Only for the RF69xR readers

3.3.3.6 getObservedTagIDs

Table 3-57 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V2.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

Please note that this command was designed for transponder processing using the RF600 readers. This command is also present with RF1xxC and RF360R for compatibility reasons, but application of it is generally not useful, except when you want to read out the UIDs of the transponders.

This command can be used in the following scenarios.

Scenario 1: No inventories are currently being performed.

In this case, you trigger inventories on the selected read points with this command. A reply frame returns information on which transponders were detected.

In contrast to the "readTagIDs" command, the smoothing algorithm of the selected read point is also applied with this command. The read point must therefore take enough inventories until a transponder adopts the "observed" status. This can be achieved by selecting the suitable parameter values or by triggering/starting the read point in good time before the command.

If no transponder with the "observed" status was detected, a positive reply is also returned but it does not contain transponder data.

To be reported, a transponder must have had the "observed" status at least once at the start of or during the command execution time. The transponder is reported even if it returned temporarily to the "lost" status.

Note that this command cannot be interrupted with RF600. The number of inventories performed and the time period over which the reader performs inventories depend on the

"value_duration" and "value_unit" parameters. An acknowledge frame from the reader is only sent after this. You must consider these reply times in the user application. The configuration parameters of the read point (read cycles per trigger, read timeout, ...) are not used. You will find more information on the parameters in the WBM sub-section "Read points" of the relevant Ident device manual.

Scenario 2: Inventories are already being performed.

In this case, the reader is already performing inventories. This can take place, for example, by means of a connected light barrier that is configured accordingly. If this command is then executed, all transponders that have the "observed" status are output.

Note

RF600: Filter mechanisms influence the results

Please note that defined filter mechanisms affect the results. With an active filter, only the transponders that have not been filtered out are listed in the reply frame.

You will find more information on the filter mechanisms in the WBM sub-section "Filters" of the relevant Ident device manual.

Note

Delay of the reply frames

Please note that the reply frame of the Ident device may be delayed further due to the configured duration of the command.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getObservedTagIDs>
        <sourceName> value_sourceName </sourceName>
        <duration> value_duration </duration>     //opt
        <unit> value_unit </unit>                 //opt
    </getObservedTagIDs>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
```

```
    <resultCode> 0 </resultCode>
    <getObservedTagIDs>
        <returnValue>
            <tag>
                <tagID> value_tagID </tagID>
                <sourceName> value_sourceName </sourceName>
                <tagPC> value_tagPC </tagPC>            //*
                <tagXPC> value_tagXPC </tagXPC>         //*
                <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                <phase> value_phase </phase>            //*
                <phaseEnd> value_phaseEnd </phaseEnd>   //*
                <utcTime> value_utcTime </utcTime>      //opt
                <antennaName> value_antennaName </antennaName>
                <rSSI> value_rSSI </rSSI>               //*
                <channel> value_channel </channel>      //*
                <power> value_power </power>            //*
                <polarization> value_polarization </polarization>  //*
                <inventoried> value_inventoried </inventoried>     //*
                <filterDataAvailable> value_filterDataAvailable
                </filterDataAvailable>                  //*
            </tag>
            ...
            <tag>        //opt
               ...
            </tag>       //opt
        </returnValue>
    </getObservedTagIDs>
  </reply>
 </frame>
```

//opt ≙ Optional: Parameter can be omitted.

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").

Parameter	Type	Values	Description
value_duration	Decimal value 0...9	0...65535	Duration for which the selected read point will read transponders. If the value "0" is set or the parameter is empty, only one read cycle is executed.
value_unit	Fixed values	Time Count	Specifies the unit for the duration. Time (default): Time in milliseconds Count: Number of inventories
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format" Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC [1)]	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [2)]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1)]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1)]	Decimal value 0...9	0...255 [2)] -10000...+10000 [3)]	RSSI value This value has no unit and is not directly related to the performance. [2)] RSSI values in 1/100 dBm [3)]
value_channel [1)]	Decimal value 0...9	1...50	Channel number on which the transponder was detected.

Parameter	Type	Values	Description
value_power [1]	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization [1]	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. "Default" ≙ "Linear_vertical".
value_inventor-ied [1]	Decimal value 0...9	0...65535	Indicates how often the transponder was identified via the air interface in this command.
value_ filterDataAvaila-ble [1]	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment

[2] Only for the RF61xR/RF650R/RF68xR readers

[3] Only for the RF69xR readers

3.3.3.7 readTagMemory

Table 3-58 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you can read out data from the selected transponder. If the specified EPC ID is not found for RF600, the command is executed for all transponders detected by the read point.

Note

RF600: Filter mechanisms may influence the results

If no EPC ID is specified, the results are influenced by all defined filter mechanisms. In this case, only the transponders that were not filtered out appear in the reply frame. If an EPC ID is specified, the results are not filtered.

You will find more information on the filter mechanisms in the WBM sub-section "Filters" of the relevant Ident device manual.

The reply frame contains the ID of the detected transponder or, with RF600, the IDs of all detected transponders, along with the information as to whether the requested data for the transponder could be read or not. If no transponder was detected, a negative reply without transponder data is returned.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <readTagMemory>
       <sourceName> value_sourceName </sourceName>
       <tagID> value_tagID </tagID>                //opt
       <password> value_password </password>       //*
       <tagField>
          <bank> value_bank </bank>               //opt
          <startAddress> value_startAddress </startAddress>
          <dataLength> value_dataLength </dataLength>
       </tagField>
       ...
       <tagField>        //opt
          ...
       </tagField>       //opt
    </readTagMemory>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <readTagMemory>
        <returnValue>
            <tag>
                <tagID> value_tagID </tagID>
                <sourceName> value_sourceName </sourceName>     //*
                <tagPC> value_tagPC </tagPC>            //*
                <tagXPC> value_tagXPC </tagXPC>         //*
                <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                <phase> value_phase </phase>            //*
                <phaseEnd> value_phaseEnd </phaseEnd>   //*
                <success> value_success </success>
                <utcTime> value_utcTime </utcTime>      //opt
                <antennaName> value_antennaName </antennaName>  //*
```

```
                <rSSI> value_rSSI </rSSI>              //*
                <channel> value_channel </channel>     //*
                <power> value_power </power>           //*
                <polarization> value_polarization </polarization>  //*
                <commandRetry> value_commandRetry </commandRetry>  //*
                <airRetry> value_airRetry </airRetry>  //*
                <filterDataAvailable> value_filterDataAvailable
                </filterDataAvailable>                 //*
                <tagField>       //opt
                  <bank> value_bank </bank>            //*
                  <startAddress> value_startAddress </startAddress>
                  <dataLength> value_dataLength </dataLength>
                  <data> value_data </data>
                </tagField>     //opt
                ...
                <tagField>      //opt
                   ...
                </tagField>     //opt
            </tag>
            ...
            <tag>        //opt
               ...
            </tag>       //opt
        </returnValue>
    </readTagMemory>
  </reply>
 </frame>
```

//opt ≙ Optional: Parameter can be omitted. In the reply, the parameter is transferred depending on the WBM parameter assignment ("Settings - Communication").

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName [1)]	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").

Parameter	Type	Values	Description
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format". This function accesses all transponders with this ID. If this parameter is empty or not even transferred, the function applies to all transponders. Nevertheless only one transponder is then permitted in the antenna field. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC [1)]	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1)]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1)]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_bank	--	0...3	RF600: Memory bank of the transponder • 0: Reserved • 1: EPC • 2: TID • 3: USER MEMORY If no value is selected, the memory bank 3 "USER MEMORY" is used automatically. RF1xxC, RF360R: Parameter not relevant. RF1100: VHL ID
value_ startAddress	Decimal value 0...9	0...4294967295	Start address of the first byte in the memory bank where reading will start.
value_ dataLength	Decimal value 0...9	1...1024	Number of bytes to be read.

Parameter	Type	Values	Description
value_password [1]	Hexadecimal value 0...9, A...F	00000000... FFFFFFFF	Access password for the transponder This parameter does not need to be defined if there is no password protection activated for the transponder.
value_success	Fixed values	True False	Indicates whether the command was successful for this transponder. True: Command successful False: Command unsuccessful
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName [1]	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
value_channel [1]	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power [1]	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization [1]	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. "Default" ≙ "Linear_vertical".
value_ commandRetry [1]	Decimal value 0...9	0...65535	Number of times the reader repeated commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of attempts is set using the WBM.
value_airRetry [1]	Decimal value 0...9	0...65535	Number of times the reader repeated air interface commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of air interface commands is specified in the reader firmware and cannot be modified.

Parameter	Type	Values	Description
value_ filterDataAvaila-ble [1]	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.
value_data	Hexadecimal value 0...9, A...F	--	Data that was read. Each byte is represented by two hexadecimal characters. Example: The byte sequence "0x12, 0x34, 0xA3" is represented as the character string "1234A3" in the "value_data" parameter. "value_dataLength" is 3 in this example. If the transponder is detected but the data cannot be read (e.g. if the transponder has no user memory as required), this field remains empty. "value_success" is then also set to "False".

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment

[2] Only for the RF61xR/RF650R/RF68xR readers

[3] Only for the RF69xR readers

3.3.3.8 writeTagMemory

Table 3-59 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

With this command, you write data to the selected transponder. If the specified EPC ID is not found for RF600, the command is executed for all transponders detected by the read point.

Note

RF600: Filter mechanisms may influence the results

If no EPC ID is specified, the results are influenced by all defined filter mechanisms. In this case, only the transponders that were not filtered out appear in the reply frame. If an EPC ID is specified, the results are not filtered.

You will find more information on the filter mechanisms in the WBM sub-section "Filters" of the relevant Ident device manual.

Note

Unaddressed write operations to memory bank "EPC"

If the optional parameter "tagID" is not specified, the reader runs a write operation on all transponders that are located within the antenna field. If the "EPC" memory bank is written in the process, this can have the result that the same EPC ID/UID is assigned to all transponders and that they can no longer be addressed individually by subsequent commands. Therefore, if possible, the "writeTagID" command should always be used instead of "writeTagMemory" to write to the "EPC" memory bank.

Note

Write operations to "EPC" memory bank with ID length "0"

When writing to the "EPC" memory bank, make sure that the length field of the EPC ID/UID is not accidentally set to "0". This has the result that the affected transponders can then no longer be addressed via the "tagID" parameter. In many cases, this amounts to a permanent total failure of the transponder. Therefore, if possible, the "writeTagID" command should always be used instead of "writeTagMemory" to write to the "EPC" memory bank.

The reply frame contains the IDs of the detected transponder or, with RF600, the IDs of all detected transponders. The "value_success" parameter indicates for each transponder whether or not the command was successful. If no transponder was identified, a negative reply is returned.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <writeTagMemory>
       <sourceName> value_sourceName </sourceName>
       <tagID> value_tagID </tagID>                //opt
       <password> value_password </password>       //opt
       <tagField>
          <bank> value_bank </bank>                //opt
          <startAddress> value_startAddress </startAddress>
          <dataLength> value_dataLength </dataLength>
          <data> value_data </data>                //opt
       </tagField>
       ...
       <tagField>        //opt
          ...
       </tagField>       //opt
    </writeTagMemory>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <writeTagMemory>
        <returnValue>
            <tag>
                <tagID> value_tagID </tagID>
                <sourceName> value_sourceName </sourceName>     //*
                <tagPC> value_tagPC </tagPC>            //*
                <tagXPC> value_tagXPC </tagXPC>         //*
                <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                <phase> value_phase </phase>            //*
                <phaseEnd> value_phaseEnd </phaseEnd>   //*
                <success> value_success </success>
                <utcTime> value_utcTime </utcTime>      //*
                <antennaName> value_antennaName </antennaName>  //*
                <rSSI> value_rSSI </rSSI>              //*
                <channel> value_channel </channel>     //*
                <power> value_power </power>           //*
                <polarization> value_polarization </polarization>  //*
                <commandRetry> value_commandRetry </commandRetry>  //*
                <airRetry> value_airRetry </airRetry>  //*
                <filterDataAvailable> value_filterDataAvailable
                </filterDataAvailable>                 //*
            </tag>
            ...
            <tag>        //opt
               ...
            </tag>       //opt
        </returnValue>
    </writeTagMemory>
 </reply>
</frame>
```

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName [1)]	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format". This function accesses all transponders with this ID. If this parameter is empty or not even transferred, the function applies to all transponders. Nevertheless only one transponder is then permitted in the antenna field. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC [1)]	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1)]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1)]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees

Parameter	Type	Values	Description
value_bank	--	0...3	RF600: Memory bank of the transponder • 0: Reserved • 1: EPC • 2: TID • 3: USER MEMORY If no value is selected, the memory bank 3 "USER MEMORY" is used automatically. RF1xxC, RF360R: Parameter not relevant. RF1100: VHL ID
value_ startAddress	Decimal value 0...9	0...4294967295	Start address of the first byte in the memory bank where writing will start.
value_ dataLength	Decimal value 0...9	1...1024	Number of bytes to be written. The reader checks the correct length of "value_data". If the check is negative, a negative reply is sent.
value_data	Hexadecimal value 0...9, A...F	--	Data that should be written. Each byte is represented by two hexadecimal characters. Example: The byte sequence "0x12, 0x34, 0xA3" is represented as the character string "1234A3" in the "value_data" parameter. "value_dataLength" is 3 in this example. If the transponder is detected but the data cannot be written (e.g. if the transponder has no user memory as required), this field remains empty. "value_success" is then also set to "False".
value_password	Hexadecimal value 0...9, A...F	00000000... FFFFFFFF	Access password for the transponder This parameter does not need to be defined if there is no password protection activated for the transponder.
value_success	Fixed values	True False	Indicates whether the command was successful for this transponder. True: Command successful False: Command unsuccessful
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna

Parameter	Type	Values	Description
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
value_channel [1]	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power [1]	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization [1]	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. "Default" ≙ "Linear_vertical".
value_ commandRetry [1]	Decimal value 0...9	0...65535	Number of times the reader repeated commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of attempts is set using the WBM.
value_airRetry [1]	Decimal value 0...9	0...65535	Number of times the reader repeated air interface commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of air interface commands is specified in the reader firmware and cannot be modified.
value_ filterDataAvaila-ble [1]	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment

[2] Only for the RF61xR/RF650R/RF68xR readers

[3] Only for the RF69xR readers

3.3.3.9 readTagField

Table 3-60 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V2.2	V1.0	V2.2	V1.0

With this command, you can read out data from the specific area of the selected transponder. The address of the data area is specified by the name of a tag field. The tag field and the

name of the field are specified using the WBM. If the specified EPC ID is not found, the command is executed for all transponders detected by the read point.

Note

RF600: Filter mechanisms may influence the results

If no EPC ID is specified, the results are influenced by all defined filter mechanisms. In this case, only the transponders that were not filtered out appear in the reply frame. If an EPC ID is specified, the results are not filtered.

You will find more information on the filter mechanisms in the WBM sub-section "Filters" of the relevant Ident device manual.

The reply frame contains the IDs of all detected transponders with the information as to whether the requested data for the transponder could be read or not. If no transponder was detected, a negative reply without transponder data is returned.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
        <readTagField>
        <sourceName> value_sourceName </sourceName>
        <tagID> value_tagID </tagID>               //opt
        <password> value_password </password>      //opt
        <tagField>
            <fieldName> value_fieldName </fieldName>
        </tagField>
       ...
       <tagField>         //opt
          ...
       </tagField>        //opt
    </readTagField>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
```

```
    <readTagField>
        <returnValue>
            <tag>
                <tagID> value_tagID </tagID>
                <tagPC> value_tagPC </tagPC>            //opt
                <tagXPC> value_tagXPC </tagXPC>         //*
                <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                <phase> value_phase </phase>            //*
                <phaseEnd> value_phaseEnd </phaseEnd>   //*
                <success> value_success </success>
                <utcTime> value_utcTime </utcTime>      //opt
                <antennaName> value_antennaName </antennaName>  //opt
                <rSSI> value_rSSI </rSSI>              //opt
                <channel> value_channel </channel>     //opt
                <power> value_power </power>           //opt
                <polarization> value_polarization </polarization>  //opt
                <commandRetry> value_commandRetry </commandRetry>  //opt
                <airRetry> value_airRetry </airRetry>  //opt
                <filterDataAvailable> value_filterDataAvailable
                </filterDataAvailable>                 //opt
                <tagField>      //opt
                    <data> value_data </data>
                </tagField>     //opt
                …
                <tagField>      //opt
                   ...
                </tagField>     //opt
            </tag>
            ...
            <tag>        //opt
               ...
            </tag>       //opt
        </returnValue>
    </readTagField>
  </reply>
 </frame>
```

//opt ≙ Optional: Parameter can be omitted. In the reply, the parameter is transferred depending on the WBM parameter assignment ("Settings - Communication").

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format". This function accesses all transponders with this ID. If this parameter is empty or not even transferred, the function applies to all transponders. Nevertheless only one transponder is then permitted in the antenna field. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1)]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1)]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_ fieldName	Hexadecimal value 0...9, A...F	--	Name of tag field Is specified using the WBM.
value_password	Hexadecimal value 0...9, A...F	00000000... FFFFFFFF	Access password for the transponder This parameter does not need to be defined if there is no password protection activated for the transponder.

Parameter	Type	Values	Description
value_success	Fixed values	True False	Indicates whether the command was successful for this transponder. True: Command successful False: Command unsuccessful
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
value_channel	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. "Default" ≙ "Linear_vertical".
value_ commandRetry	Decimal value 0...9	0...65535	Number of times the reader repeated commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of attempts is set using the WBM.
value_airRetry	Decimal value 0...9	0...65535	Number of times the reader repeated air interface commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of air interface commands is specified in the reader firmware and cannot be modified.

Parameter	Type	Values	Description
value_ filterDataAvailable	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.
value_data	Hexadecimal value 0...9, A...F	--	Data that should be read. Each byte is represented by two hexadecimal characters. Example: The byte sequence "0x12, 0x34, 0xA3" is represented as the character string "1234A3" in the "value_data" parameter. "value_dataLength" is 3 in this example. If the transponder is detected but the data cannot be read (e.g. if the transponder has no user memory as required), this field remains empty. "value_success" is then also set to "False".

1) Depending on the Ident device used, the FW version or the WBM parameter assignment

2) Only for the RF61xR/RF650R/RF68xR readers

3) Only for the RF69xR readers

3.3.3.10 writeTagField

Table 3-61 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V2.2	V1.0	V2.2	V1.0

With this command, you write data to the specific area of the transponder. The address of the data area is specified by the name of a tag field. The tag field and the name of the field are specified using the WBM. If the specified EPC ID is not found, the command is executed for all transponders detected by the read point.

Note

RF600: Filter mechanisms may influence the results

If no EPC ID is specified, the results are influenced by all defined filter mechanisms. In this case, only the transponders that were not filtered out appear in the reply frame. If an EPC ID is specified, the results are not filtered.

You will find more information on the filter mechanisms in the WBM sub-section "Filters" of the relevant Ident device manual.

Note

Unaddressed write operations to memory bank "EPC"

If the optional parameter "tagID" is not specified, the reader runs a write operation on all transponders that are located within the antenna field. If the "EPC" memory bank is written in the process, this can have the result that the same EPC ID/UID is assigned to all transponders and that they can no longer be addressed individually by subsequent commands. Therefore, if possible, the "writeTagID" command should always be used instead of "writeTagField" to write to the "EPC" memory bank.

Note

Write operations to "EPC" memory bank with ID length "0"

When writing to the "EPC" memory bank, make sure that the length field of the EPC ID/UID is not accidentally set to "0". This has the result that the affected transponders can then no longer be addressed via the "tagID" parameter. In many cases, this amounts to a permanent total failure of the transponder. Therefore, if possible, the "writeTagID" command should always be used instead of "writeTagField" to write to the "EPC" memory bank.

The reply frame contains the IDs of all identified transponders. The "value_success" parameter indicates for each transponder whether or not the command was successful. If no transponder was identified, a negative reply is returned.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
        <writeTagField>
        <sourceName> value_sourceName </sourceName>
        <tagID> value_tagID </tagID>               //opt
        <password> value_password </password>      //opt
        <tagField>
            <fieldName> value_fieldName </fieldName>
            <data> value_data </data>
        </tagField>
       ...
       <tagField>         //opt
          ...
       </tagField>        //opt
    </writeTagField>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <writeTagField>
        <returnValue>
            <tag>
                <tagID> value_tagID </tagID>
                <tagPC> value_tagPC </tagPC>            //opt
                <tagXPC> value_tagXPC </tagXPC>         //*
                <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                <phase> value_phase </phase>            //*
                <phaseEnd> value_phaseEnd </phaseEnd>   //*
                <success> value_success </success>
                <utcTime> value_utcTime </utcTime>      //opt
                <antennaName> value_antennaName </antennaName>  //opt
                <rSSI> value_rSSI </rSSI>              //opt
                <channel> value_channel </channel>     //opt
                <power> value_power </power>           //opt
                <polarization> value_polarization </polarization>  //opt
                <commandRetry> value_commandRetry </commandRetry>  //opt
                <airRetry> value_airRetry </airRetry>  //opt
                <filterDataAvailable> value_filterDataAvailable
                </filterDataAvailable>                 //opt
            </tag>
            ...
            <tag>        //opt
               ...
            </tag>       //opt
        </returnValue>
    </writeTagField>
 </reply>
</frame>
```

//opt ≙ Optional: Parameter can be omitted. In the reply, the parameter is transferred depending on the WBM parameter assignment ("Settings - Communication").

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format". This function accesses all transponders with this ID. If this parameter is empty or not even transferred, the function applies to all transponders. Nevertheless only one transponder is then permitted in the antenna field. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1)]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1)]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_ fieldName	Hexadecimal value 0...9, A...F	--	Name of tag field Is specified using the WBM.
value_password	Hexadecimal value 0...9, A...F	00000000... FFFFFFFF	Access password for the transponder This parameter does not need to be defined if there is no password protection activated for the transponder.

Parameter	Type	Values	Description
value_data	Hexadecimal value 0...9, A...F	--	Data that should be written. Each byte is represented by two hexadecimal characters. Example: The byte sequence "0x12, 0x34, 0xA3" is represented as the character string "1234A3" in the "value_data" parameter. "value_dataLength" is 3 in this example. If the transponder is detected but the data cannot be written (e.g. if the transponder has no user memory as required), this field remains empty. "value_success" is then also set to "False".
value_success	Fixed values	True False	Indicates whether the command was successful for this transponder. True: Command successful False: Command unsuccessful
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1)]	Decimal value 0...9	0...255 [2)] -10000...+10000 [3)]	RSSI value This value has no unit and is not directly related to the performance. [2)] RSSI values in 1/100 dBm [3)]
value_channel	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. "Default" ≙ "Linear_vertical".
value_ commandRetry	Decimal value 0...9	0...65535	Number of times the reader repeated commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of attempts is set using the WBM.

Parameter	Type	Values	Description
value_airRetry	Decimal value 0...9	0...65535	Number of times the reader repeated air interface commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of air interface commands is specified in the reader firmware and cannot be modified.
value_ filterDataAvailable	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.

1) Depending on the Ident device used, the FW version or the WBM parameter assignment

2) Only for the RF61xR/RF650R/RF68xR readers

3) Only for the RF69xR readers

3.3.3.11 killTag

Table 3-62 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	--	--	--	--

With this command, you disable the selected transponder. If the specified EPC ID is not found, the command is executed for all transponders detected by the read point.

Note

RF600: Filter mechanisms may influence the results

If no EPC ID is specified, the results are influenced by all defined filter mechanisms. In this case, only the transponders that were not filtered out appear in the reply frame. If an EPC ID is specified, the results are not filtered.

You will find more information on the filter mechanisms in the WBM sub-section "Filters" of the relevant Ident device manual.

The reply frame contains the IDs of all identified transponders. The "value_success" parameter indicates for each transponder whether or not the command was successful. If no transponder was identified, a negative reply is returned.

Command

```
<frame>
 <cmd>
```

```
    <id> value_id </id>
    <killTag>
        <sourceName> value_sourceName </sourceName>
        <tagID> value_tagID </tagID>              //opt
        <password> value_password </password>
    </killTag>
  </cmd>
 </frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
 <frame>
  <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <killTag>
        <returnValue>
            <tag>
                <tagID> value_tagID </tagID>
                <tagPC> value_tagPC </tagPC>            //opt
                <tagXPC> value_tagXPC </tagXPC>         //*
                <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                <phase> value_phase </phase>            //*
                <phaseEnd> value_phaseEnd </phaseEnd>   //*
                <success> value_success </success>
                <utcTime> value_utcTime </utcTime>      //opt
                <antennaName> value_antennaName </antennaName>  //opt
                <rSSI> value_rSSI </rSSI>              //opt
                <channel> value_channel </channel>     //opt
                <power> value_power </power>           //opt
                <polarization> value_polarization </polarization>  //opt
                <commandRetry> value_commandRetry </commandRetry>  //opt
                <airRetry> value_airRetry </airRetry>  //opt
                <filterDataAvailable> value_filterDataAvailable
                </filterDataAvailable>                 //opt
            </tag>
            ...
            <tag>        //opt
               ...
            </tag>       //opt
        </returnValue>
    </killTag>
```

```
  </reply>
 </frame>
```

//opt ≙ Optional: Parameter can be omitted. In the reply, the parameter is transferred depending on the configuration settings (engineering/communication).

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format". This function accesses all transponders with this ID. If this parameter is empty or not even transferred, the function applies to all transponders. Nevertheless only one transponder is then permitted in the antenna field. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1)]	Decimal value 0...9	0...360	Phase information in degrees

Parameter	Type	Values	Description
value_phaseEnd [1]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_password	Hexadecimal value 0...9, A...F	00000000... FFFFFFFF	Disable the kill password of the transponder
value_success	Fixed values	True False	Flag to indicate whether or not the command was successful for this transponder. True: Command successful False: Command unsuccessful
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
value_channel	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. "Default" ≙ "Linear_vertical".
value_ commandRetry	Decimal value 0...9	0...65535	Number of times the reader repeated commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of attempts is set using the WBM.

Parameter	Type	Values	Description
value_airRetry	Decimal value 0...9	0...65535	Number of times the reader repeated air interface commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of air interface commands is specified in the reader firmware and cannot be modified.
value_ filterDataAvailable	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.

1) Depending on the Ident device used, the FW version or the WBM parameter assignment

2) Only for the RF61xR/RF650R/RF68xR readers

3) Only for the RF69xR readers

3.3.3.12 lockTagBank

Table 3-63 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	--	--	--	--

With this command, you lock or unlock the corresponding memory area of the selected transponder. If the specified EPC ID is not found, the command is executed for all transponders detected by the read point.

Note

RF600: Filter mechanisms may influence the results

If no EPC ID is specified, the results are influenced by all defined filter mechanisms. In this case, only the transponders that were not filtered out appear in the reply frame. If an EPC ID is specified, the results are not filtered.

You will find more information on the filter mechanisms in the WBM sub-section "Filters" of the relevant Ident device manual.

The reply frame contains the IDs of all identified transponders. The "value_success" parameter indicates for each transponder whether or not the command was successful. If no transponder was identified, a negative reply is returned.

Structure of the parameters "epcGen2LockAction" and "epcGen2LockMask"

The first row of the table ('Bit') indicates the bit positions of the action and the masking values. The masking and action values are specified first with the MSB.

Further information on the parameters can be found in the "EPCglobal Specification (http://www.gs1.org)".

Table 3-64 Structure of the parameters "epcGen2LockAction" and "epcGen2LockMask"

Bank	Kill PWD		Access PWD		EPC Memory		TID Memory		User Memory	
Bit	9	8	7	6	5	4	3	2	1	0
Masking	s/w	s/w	s/w	s/w	s/w	s/w	s/w	s/w	s/w	s/w
Action	r/w	p	r/w	p	w	p	w	p	w	p

Explanations on the table:

- The masking value specifies which action value bit is to be set as s/w as shown in the table above ("skip/write" with "skip=0" and "write=1").
- The action value specifies which lock should be defined for each memory bank.
  - "r/w" locks the bank for read and write access.
  - "w" locks the bank for write access (read access permitted).
  - "p" specifies a permanent lock.

The following tables show possible combinations of the "r/w" and "w" action values with/without lock ("p") and the respective meaning for a memory bank.

The transponder is in the "open" status when it is identified and in the "secured" status when its access password is verified.

Table 3-65 Possible combinations of the "w" action values

w	p	Description
0	0	Write access to the particular memory bank is possible from both the "open" and the "secured" status.
0	1	Write access to the particular memory bank is permanently possible from both the "open" and the "secured" status; it can never be locked.
1	0	Write access to the particular memory bank is possible from the "secured" status but not from the "open" status.
1	1	No write access to the particular memory bank is possible.

Table 3-66 Possible combinations of the "r/w" action values

r/w	p	Description
0	0	Read and write access to the particular password is possible from both the "open" and the "secured" status.
0	1	Read and write access to the particular password is permanently possible from both the "open" and the "secured" status; it can never be locked.
1	0	Read and write access to the particular password is possible from the "secured" status but not from the "open" status.
1	1	No read or write access to the particular memory bank is possible.

Example

Bank	Kill PWD	Access PWD	EPC Memory	TID Memory	User Memory	Hex string value
Masking	(00) 11	11	11	00	00	3F0
Action	(00) 10	10	10	00	00	2A0

In this example, the masking value is "lockMask = 11 1111 0000" (hex: 3F0). As a result, only write access to the memory locations "Kill", "Access" and "EPC" is possible. The action value "lockAction = 10 1010 0000" (hex: 2A0) has the following effects:

- Kill password
  Read and write access is possible from the "secured" status, but not from the "open" status. The Access password for the transponder must be known before the Kill password can be read or modified.
- EPC memory bank
  Write access is possible from the "secured" status, but not from the "open" status. The Access password must be known before a new ID can be written to the transponder. You will find more detailed information on "epcGen2LockAction" and "epcGen2LockMask" in the "EPC Radio Frequency Identity Protocols Standard Specification".

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <lockTagBank>
        <sourceName> value_sourceName </sourceName>
        <tagID> value_tagID </tagID>              //opt
        <lockAction> value_lockAction </lockAction>
        <lockMask> value_lockMask </lockMask>
        <password> value_password </password>
    </lockTagBank>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <lockTagBank>
        <returnValue>
```

```
                <tag>
                    <tagID> value_tagID </tagID>
                    <tagPC> value_tagPC </tagPC>            //opt
                    <tagXPC> value_tagXPC </tagXPC>         //*
                    <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                    <phase> value_phase </phase>            //*
                    <phaseEnd> value_phaseEnd </phaseEnd>   //*
                    <success> value_success </success>
                    <utcTime> value_utcTime </utcTime>      //opt
                    <antennaName> value_antennaName </antennaName>  //opt
                    <rSSI> value_rSSI </rSSI>              //opt
                    <channel> value_channel </channel>     //opt
                    <power> value_power </power>           //opt
                    <polarization> value_polarization </polarization>  //opt
                    <commandRetry> value_commandRetry </commandRetry>  //opt
                    <airRetry> value_airRetry </airRetry>  //opt
                    <filterDataAvailable> value_filterDataAvailable
                    </filterDataAvailable>                 //opt
                </tag>
                ...
                <tag>        //opt
                   ...
                </tag>       //opt
            </returnValue>
        </lockTagBank>
    </reply>
</frame>
```

//opt ≙ Optional: Parameter can be omitted. In the reply, the parameter is transferred depending on the configuration settings (engineering/communication).

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").

Parameter	Type	Values	Description
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format". This function accesses all transponders with this ID. If this parameter is empty or not even transferred, the function applies to all transponders. Nevertheless only one transponder is then permitted in the antenna field. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1)]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1)]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1)]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_ lockAction	Boolean value 0, 1	0000000000... 1111111111	"LockAction" is a 10-digit Boolean value. The least significant bits decide the "USER Memory". You will find a full description of the "LockAction" parameter in the "EPC Global Specification".
value_lockMask	Boolean value 0, 1	0000000000... 1111111111	"LockMask" is a 10-digit Boolean value. The least significant bits decide the "USER Memory". You will find a full description of the "LockMask" parameter in the "EPC Global Specification".
value_password	Hexadecimal value 0...9, A...F	00000000.... FFFFFFFF	Access password for the transponder

Parameter	Type	Values	Description
value_success	Fixed values	True False	Flag to indicate whether or not the command was successful for this transponder. True: Command successful False: Command unsuccessful
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
value_channel	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization	Fixed values	Default Circular Linear_vertical Linear_horizontal All	Polarization of the antenna This parameter is only relevant with switchable antennas. "Default" ≙ "Linear_vertical".
value_ commandRetry	Decimal value 0...9	0...65535	Number of times the reader repeated commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of attempts is set using the WBM.
value_airRetry	Decimal value 0...9	0...65535	Number of times the reader repeated air interface commands attempting to obtain correct data. "0" corresponds to no attempt. The maximum number of air interface commands is specified in the reader firmware and cannot be modified.
value_ filterDataAvailable	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment

[2] Only for the RF61xR/RF650R/RF68xR readers

[3] Only for the RF69xR readers

3.3.3.13 getTagStatus

Table 3-67 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	--	--	V1.3	V1.0	V1.0	--

With this command, you can read out status information of a transponder. To ensure clear identification, either there can be only one transponder in the antenna field or the ID of the transponder needs to be specified. If there is more than one transponder in the antenna field and no transponder ID is specified, a negative response is returned.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
    <getTagStatus>
       <sourceName> value_sourceName </sourceName>
       <mode> value_mode </mode>                  //opt
       <tagID> value_tagID </tagID>               //opt
    </getTagStatus>
 </cmd>
</frame>
```

//opt ≙ Optional: Parameter can be omitted.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <getTagStatus>
        <returnValue>
           <tagStatusData> value_tagStatusData </tagStatusData>
           <tagID> value_tagID </tagID>         //opt
           <values>                             //opt
              <value> value_value </value>
              ...
           </values>
        </returnValue>
    </getTagStatus>
```

```
  </reply>
 </frame>
```

//opt ≙ Optional: Parameter can be omitted.

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_mode	Fixed values	1 2 3 239	Mode of the status data to be requested. The supported values depend on the reader type assigned to the specified read point. If this parameter is not set, the value "3" is used automatically.
value_tagID	Hexadecimal value 0...9, A...F	--	UID in "RAW Hex Data Format". This function accesses the transponder with this ID. Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_ tagStatusData	Hexadecimal value 0...9, A...F	--	The raw reply data of the "getTagStatus" command as simple byte sequence. It is mode-specific, reader-specific and transponder-specific.
value_value	Alphanumeric text	--	Named specific values from the raw reply data. Note that the number of values can change. There may be more or fewer values in future versions.

3.3.3.14 stopCommand

Table 3-68 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	V4.0	V1.0	V2.0	V1.0	V2.0	V1.0

With this command, the currently running command of the reader connected to the read point is canceled.

Command

```
<frame>
 <cmd>
    <id> value_id </id>
       <sourceName> value_sourceName </sourceName>
    <stopCommand/>
 </cmd>
</frame>
```

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <stopCommand/>
 </reply>
</frame>
```

The error codes for negative replies (resultCode ≠ 0) are described in the section "Error messages (negative XML replies) (Page 128)".

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").

3.3.4 Error messages (negative XML replies)

Table 3-69 Protocol version and firmware versions of the command

XML API	RF600	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

If a command cannot be executed, the user application receives an error message (negative XML message) from the affected device. This reply frame always contains the corresponding result code ("value_resultCode ≠ 0" or "≠ 0x00"), as well as information on the error that has occurred.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> value_resultCode </resultCode>
    <error>
        <name> value_name </name>
        <cause> value_cause </cause>
    </error>
 </reply>
</frame>
```

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ resultCode	Decimal value 0...9	0...65535	Result code of the command You will find a list of possible return codes in the following table.
value_name	Text	--	Textual description of the error You will find descriptions of the problems in the following table.
value_cause	Text	--	Brief description of the cause of the negative reply. The device causing this ("DeviceXX") may also be specified in the description. "Device00" refers here to the basic device (communications module, RF600 or RF360R). "Device01"... "Device04" refers to the device connected to the respective interface of the communications module.

List of possible result codes

The result codes specified below differ in the number space from the possible error numbers in the logbook. The error numbers in the logbook correspond to the error numbers of the Ident instructions. You can find the assignment of the result code to the Ident instruction error numbers in the "Ident instructions" column.

Error code Ident instructions	Result code ("value_ resultCode")		Name ("value_name")	Description ("value_cause")
hex	hex	dec		
--	0x00	0	NO_ERROR	Positive reply. Command processed successfully.
0xE1FE0100	0x11	17	ERROR_MEMORY_ERROR	Memory error of the transponder

Error code Ident instructions	Result code ("value_resultCode")		Name ("value_name")	Description ("value_cause")
hex	hex	dec		
0xE1FE0200	0x12	18	ERROR_PRESENCE_ERROR	Presence error The transponder left the antenna field.
0xE1FE0300	0x13	19	ERROR_ADDRESS_ERROR	Address error Access outside the addressed memory area.
0xE1FE0400	0x14	20	ERROR_TAG_DETECTIVE	Initialization error The transponder is unable to execute the initialization command.
0xE1FE0500	0x15	21	ERROR_TAG_MEMORY_OVERFLOW	Access outside the addressed memory area.
0xE1FE0600	0x16	22	ERROR_TAG_UNFORMATTED	The transponder is not formatted.
0xE1FE0700	0x17	23	ERROR_DATA_STRUCTURE	The transponder data is inconsistent.
0xE1FE0800	0x18	24	ERROR_WRONG_EPCID	The transponder does not have the expected EPC ID/UID or has no EPC ID/UID.
0xE1FE0900	0x19	25	ERROR_AIR_COMMAND_NOT_SUPPORTED	The transponder does not support the command of the wireless protocol.
0xE1FE0A00	0x1A	26	ERROR_TAG_LOCKED	The transponder to which data is to be written or which should be deactivated is locked.
0xE1FE8100	0x91	145	ERROR_NO_ASWER_FROM_TAG	The transponder is not responding.
0xE1FE8200	0x92	146	ERROR_WRONG_PASSWORD	The password you entered is incorrect. Access to the transponder was denied.
0xE1FE8300	0x93	147	ERROR_VERIFY_TAG_FAILED	Verification of the transponder failed.
0xE1FE8400	0x94	148	ERROR_TAG_UNSPECIFIED	General transponder error
0xE1FE8500	0x95	149	ERROR_TAG_INSUFFICIENT_POWER	There is not enough transponder power.
0xE2FE0100	0x21	33	ERROR_FIELD_DISTURBED	Field error or send error
0xE2FE0200	0x22	34	ERROR_TOO_MANY_TAGS	There are too many transponders in the antenna field.
0xE2FE8100	0xA1	161	ERROR_NO_TAG	There is no transponder with the relevant EPC ID/UID in the antenna field.
0xE2FE8200	0xA2	162	ERROR_NO_DATA	The requested data is not available.
0xE2FE8300	0xA3	163	ERROR_INVALID_CRC	Bad checksum
0xE2FE8500	0xA5	165	ERROR_NO_FREQUENCY	No radio channel activated.
0xE2FE8600	0xA6	166	ERROR_NO_CARRIER	No carrier signal activated.
0xE2FE8700	0xA7	167	ERROR_MORE_THAN_ONE_TAG_IN_FIELD	There is more than one transponder in the antenna field.
0xE2FE8800	0xA8	168	ERROR_AIR_PROTOCOL_UNSPECIFIED	General radio protocol error
0xE4FE0100	0x41	65	ERROR_POWER_SUPPLY	Fault in power supply

Error code Ident instructions	Result code ("value_resultCode")		Name ("value_name")	Description ("value_cause")
hex	hex	dec		
0xE4FE0300	0x43	67	ERROR_ANTENNA	An antenna error was detected when executing the command.
0xE4FE0500	0x45	69	ERROR_DATA_ PUFFER_OVERFLOW	Data buffer overflow
0xE4FE0600	0x46	70	ERROR_INVALID_ READER_STATUS	The specified command is not permitted in the current reader status.
0xE4FE0700	0x47	71	ERROR_STARTUP_ MESSAGE	Incorrect startup message
0xE4FE8100	0xC1	193	ERROR_TAGFIELD_ NOT_FOUND	The specified tag field is unknown.
0xE4FE8A00	0xCA	202	ERROR_MISCELLANEOUS	A general error has occurred.
0xE4FE8A00	0xFA7	4007	ERROR_READPOINT_NOT_FOUND	Unknown read point
0xE4FE8B00	0xCB	203	ERROR_CONFIGURATION	A configuration error has occurred.
0xE4FE8C00	0xCC	204	ERROR_PIB_READER	Communication error between Ident profile and reader/CM.
0xE4FE8D00	0xCD	205	ERROR_FIRMWARE_ INTERNAL	Internal firmware error
0xE4FE8E00	0xCE	206	ERROR_COMMAND_ ABORTED	The command was aborted.
0xE4FE8F00	0xCF	207	ERROR_WRONG_ MODULE_ CONFIGURATION	Wrong module configuration
0xE4FE9300	0x1F5	501	ERROR_ANTENNA_1	Antenna 1 not connected
0xE4FE9300	0x1F6	502	ERROR_ANTENNA_2	Antenna 2 not connected
0xE4FE9300	0x1F7	503	ERROR_ANTENNA_3	Antenna 3 not connected
0xE4FE9300	0x1F8	504	ERROR_ANTENNA_4	Antenna 4 not connected
0xE4FE9300	0x1F9	505	ERROR_ANTENNA_5	Antenna 5 not connected
0xE4FE9300	0x1FA	506	ERROR_ANTENNA_6	Antenna 6 not connected
0xE4FE9300	0x1FB	507	ERROR_ANTENNA_7	Antenna 7 not connected
0xE4FE9300	0x1FC	508	ERROR_ANTENNA_8	Antenna 8 not connected
0xE4FE9300	0x1FD	509	ERROR_WRONG_TYPE_ OR_VERSION_ANTENNA_1	Antenna 1: Antenna type or version is incorrect.
0xE4FE9300	0x1FE	510	ERROR_WRONG_TYPE_ OR_VERSION_ANTENNA_2	Antenna 2: Antenna type or version is incorrect.
0xE4FE9300	0x1FF	511	ERROR_WRONG_TYPE_ OR_VERSION_ANTENNA_3	Antenna 3: Antenna type or version is incorrect.
0xE4FE9300	0x200	512	ERROR_WRONG_TYPE_ OR_VERSION_ANTENNA_4	Antenna 4: Antenna type or version is incorrect.
0xE4FE9300	0x201	513	ERROR_WRONG_TYPE_ OR_VERSION_ANTENNA_5	Antenna 5: Antenna type or version is incorrect.
0xE4FE9300	0x202	514	ERROR_WRONG_TYPE_ OR_VERSION_ANTENNA_6	Antenna 6: Antenna type or version is incorrect.
0xE4FE9300	0x203	515	ERROR_WRONG_TYPE_ OR_VERSION_ANTENNA_7	Antenna 7: Antenna type or version is incorrect.

Error code Ident instructions	Result code ("value_resultCode")		Name ("value_name")	Description ("value_cause")
hex	hex	dec		
0xE4FE9300	0x204	516	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_8	Antenna 8: Antenna type or version is incorrect.
0xE5FE8100	0xD1	209	ERROR_COMMUNICATION_ACCESS_DENIED	Communication error between CM and reader or internal communication error to the RFID module: Access denied.
0xE5FE8200	0xD2	210	ERROR_COMMUNICATION_RESSOURCE_USED	Communication error between CM and reader or internal communication error to the RFID module: Resource occupied.
0xE5FE8300	0xD3	211	ERROR_COMMUNICATION_SERIAL_INTERFACE	Functional error of the serial interface
0xE5FE8400	0xD4	212	ERROR_COMMUNICATION_COMMON_ERROR	Communication error between CM and reader: General error
0xE6FE0100	0x61	97	ERROR_COMMAND_NOT_SUPPORTED	The command is not supported by the reader or is unknown.
0xE6FE0200	0x62	98	ERROR_INVALID_COMMAND_INDEX	Invalid command index (CI)
0xE6FE0300	0x63	99	ERROR_PARAMETER_INVALID_VALUE	A parameter has an invalid value.
0xE6FE0400	0x64	100	ERROR_TAG_CROSSED_FIELD_WITHOUT_CMD_PROCESSED	A transponder has passed through the antenna field without being processed.
0xE6FE0500	0xE1	225	ERROR_PARAMETER_MISSING	A necessary parameter has not been specified.
0xE6FE8100	0xE2	226	ERROR_PARAMETER_INVALID_FORMAT	A parameter has an incorrect format
0xE6FE8200	0xE3	227	ERROR_PARAMETER_INVALID_TYPE	A parameter has the wrong format or the wrong data type.
0xE6FE8300	0xE4	228	ERROR_PARAMETER_NOT_SUPPORTED	A parameter is not supported by this reader.
0xE6FE8400	0xE5	229	ERROR_WRONG_MESSAGE_FORMAT	The XML format is incorrect. An error was detected when parsing the command.
0xE6FE8600	0xE6	230	ERROR_INVENTORY_COMMAND_FAILED	The "Inventory" command failed.
0xE6FE8700	0xE7	231	ERROR_READ_COMMAND_FAILED	The read command failed.
0xE6FE8800	0xE8	232	ERROR_WRITE_COMMAND_FAILED	The write command failed.
0xE6FE8900	0xE9	233	ERROR_WRITETAGID_COMMAND_FAILED	Writing the EPC-ID failed.
0xE6FE8A00	0xEA	234	ERROR_LOCK_COMMAND_FAILED	The "Lock" command failed.
0xE6FE8B00	0xEB	235	ERROR_KILL_COMMAND_FAILED	The "Kill" command failed.
0xE6FE8C00	0xEC	236	ERROR_STOP_COMMAND_FAILED	The "Stop" command failed.

3.4 XML reports

This section describes all frames that can be sent automatically by the connected device to the user application (XML reports). These reports can be transferred transacted or not transacted and are differentiated based on their type - event or alarm. You will find more information on XML reports in the section "Principle of operation (Page 8)".

Note

Displayed parameters and optional rows

The following sections explain the XML commands and responses, as well as reports in detail. Note that all parameters that can theoretically be displayed will be described. The availability of the parameters depends on the Ident device used, the firmware version or the parameter assignment in the WBM. These parameters are designated by the code "*".

Some of the parameters are optional: These parameters are designated by the code "opt".

3.4.1 Events

This section describes all events. Events are asynchronous frames sent automatically by the device to the user application and containing data acquired by the reader itself.

3.4.1.1 tagEventReport

Table 3-70 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	V1.0	V1.0	V1.0

On a status change (glimpsed, observed, lost) of a transponder, a transponder event report records the transponder data and sends it to the user application.

With RF600, the configuration of the trigger (continuous, I/O level, ...) and the definition of the transponder data (tag fields, RSSI value, ...) to be sent are taken from the stored configuration. These parameters cannot only be changed using the WBM. The complete transponder data is only transferred with an Observed event.

The transponder data in each event report is grouped according to read points (<source>).

A reply frame from the user application is necessary only when transacted transfer is set. A reply frame can, however, also be sent in non-secure mode and this has no negative influence.

Report

```
<frame>
```

```
<report>
   <id> value_id </id>
   <ter>
       <source>
           <sourceName> value_sourceName </sourceName>
           <tag>
               <tagID> value_tagID </tagID>
               <sourceName> value_sourceName </sourceName>
               <tagPC> value_tagPC </tagPC>                //*
               <tagXPC> value_tagXPC </tagXPC>         //*
               <tagXPC2> value_tagXPC2 </tagXPC2>      //*
               <phase> value_phase </phase>            //*
               <phaseEnd> value_phaseEnd </phaseEnd>   //*
               <event> value_event </event>                //opt
               <utcTime> value_utcTime </utcTime>          //opt
               <antennaName> value_antennaName </antennaName>
               <rSSI> value_rSSI </rSSI>                   //*
               <rSSIMin> value_rSSIMin </rSSIMin>              //*
               <rSSIMax> value_rSSIMax </rSSIMax>              //*
               <channel> value_channel </channel>          //*
               <power> value_power </power>                //*
               <polarization> value_polarization </polarization> //*
               <inventoried> value_inventoried </inventoried>  //*
               <filterDataAvailable> value_filterDataAvailable
               </filterDataAvailable>                          //*
               <tagField>            //opt
                   <fieldName> value_fieldName </fieldName>
                   <bank> value_bank </bank>
                   <startAddress> value_startAddress </startAddress>
                   <dataLength> value_dataLength </dataLength>
                   <data> value_data </data>
               </tagField>     //opt
               <tagField>      //opt
                  ...
               </tagField>     //opt
           </tag>
           ...
           <tag>       //opt
              ...
           </tag>      //opt
       </source>
       <source>        //opt
       ...
       </source>      // opt
```

```
    </ter>
  </report>
 </frame>
```

//opt ≙ Optional: The number of parameters varies depending on device configuration and/or device type. Multiple Tag events can be reported in a frame.

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

Reply

```
 <frame>
  <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <ter/>
  </reply>
 </frame>
```

A reply frame from the user application is optional. It is only necessary when transacted mode is set.

No negative replies are defined for reports.

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format" Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_tagPC [1)]	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".

Parameter	Type	Values	Description
value_tagXPC [1]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_event	Fixed values	Glimpsed [1] Observed Lost	The transponder events are also generated by a smoothing algorithm in the read point. The algorithms are specified in the WBM. A report can also contain more than one event for the same transponder. Each individual event has its own transponder structure. If a report contains two events for the same transponder, two transponder structures exist with the same "value_tagID" but different values for "value_event".
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
rSSIMin [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	The lowest RSSI value with which the transponder was detected. This value has no unit and is not directly related to the performance. [3] RSSI values in 1/100 dBm [2]
rSSIMax [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	The highest RSSI value with which the transponder was detected. This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]

Parameter	Type	Values	Description
value_channel [1]	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power [1]	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization [1]	Fixed values	Circular Linear_vertical Linear_horizontal Unknown	Polarization of the antenna With external antennas always unknown.
value_ inventoried [1]	Decimal value 0...9	0...65535	Indicates how often the transponder was identified via the air interface before it changed to the "observed" status.
value_ filterDataAvailable [1]	Fixed values	True False	Indicates whether a filter criterion was received. True: All the data was read or no filter was set. False: Data could not be read.
value_ fieldName [1]	Text	--	Name of tag field Is specified in the WBM.
value_bank	--	0...3	Memory bank of the transponder • 0: Reserved • 1: EPC • 2: TID • 3: USER MEMORY
value_ startAddress	Decimal value 0...9	0...65535	Start address of the first byte in the memory bank where reading will start.
value_ dataLength	Decimal value 0...9	1...510	Number of bytes to be read.
value_data	Hexadecimal value 0...9, A...F	--	Data that should be read. Each byte is represented by two hexadecimal characters. Example: The byte sequence "0x12, 0x34, 0xA3" is represented as the character string "1234A3" in the "value_data" parameter. "value_dataLength" is 3 in this example. If the transponder is detected but the data cannot be read (e.g. if the transponder has no user memory as required), this field remains empty. "value_success" is then also set to "False".

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment

[2] Only for the RF61xR/RF650R/RF68xR readers

[3] Only for the RF69xR readers

3.4.1.2 rssiEventReport

Table 3-71 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	--	--	--	--

An RSSI event report signals every change in the RSSI values when reading the transponders. RSSI events are generated separately for each antenna.

If an RSSI value of 0 is output in RF600, it can have the following causes:

- If a transponder that was detected by an antenna in the last read operation is no longer detected in the subsequent read operation, the RSSI value "0" will be output once.
- If an antenna fails to detect any transponders during a read operation, the RSSI value "0" is output, and no ID is displayed in the parameter "value_tagID".

Which parameters are output is specified by the settings in the configuration in the WBM.

Report

```
<frame>
 <report>
    <id> value_id </id>
    <rssier>
        <tag>
           <tagID> value_tagID </tagID>
           <sourceName> value_sourceName </sourceName>
           <tagPC> value_tagPC </tagPC>            //*
           <tagXPC> value_tagXPC </tagXPC>         //*
           <tagXPC2> value_tagXPC2 </tagXPC2>      //*
           <phase> value_phase </phase>            //*
           <phaseEnd> value_phaseEnd </phaseEnd>   //*
           <utcTime> value_utcTime </utcTime>      //*
           <antennaName> value_antennaName </antennaName>     //*
           <rSSI> value_rSSI </rSSI>               //*
           <channel> value_channel </channel>      //*
           <power> value_power </power>            //*
           <polarization> value_polarization </polarization>  //*
        </tag>
        ...
        <tag>       //opt
           ...
        </tag>      //opt
    </rssier>
```

```
  </report>
 </frame>
```

//opt ≙ Optional: The number of parameters varies depending on device configuration and/or device type. Multiple RSSI events can be reported in a frame.

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

Reply

```
 <frame>
  <reply>
     <id> value_id </id>
     <resultCode> 0 </resultCode>
     <rssier/>
  </reply>
 </frame>
```

A reply frame from the user application is optional. It is only necessary when transacted mode is set.

No negative replies are defined for reports.

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format" Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagPC	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexa-decimal characters. Example: The value "1234" corresponds to the bi-nary value "0001.0010.0011.0100".

Parameter	Type	Values	Description
value_tagXPC [1]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
value_channel	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization	Fixed values	Circular Linear_vertical Linear_horizontal Unknown	Polarization of the antenna With external antennas always unknown.

[1] Depending on the Ident device used, the FW version or the WBM parameter assignment

[2] Only for the RF61xR/RF650R/RF68xR readers

[3] Only for the RF69xR readers

3.4.1.3 ioEventReport

Table 3-72 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V1.0	V1.0	V1.0	V1.3	--	--	--

An IO event report signals changes at a digital input/output.

The sending of IO events can be configured in the WBM. You will find more information on the events in the WBM sub-section "Communication" of the relevant Ident device manual.

Report

```
<frame>
 <report>
    <id> value_id </id>
    <ioer>
        <io>
            <ioName> value_ioName </ioName>
            <ioEvent> value_ioEvent </ioEvent>
            <utcTime> value_utcTime </utcTime>
        </io>
        <io>      //opt
        ...
        </io>     //opt
    </ioer>
 </report>
</frame>
```

//opt ≙ Optional: Multiple "IO events" can be reported in a frame.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <ioer/>
 </reply>
</frame>
```

A reply frame from the user application is optional. It is only necessary when transacted mode is set.

No negative replies are defined for reports.

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ioName	Fixed values	Inport00 Inport01 ... Inport08 Outport00 Outport01 ... Outport08	Name of the digital input/output
value_ioEvent	Fixed values	High Low	Indicates the new status of the digital input/output.
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.

3.4.1.4 logEventReport

Table 3-73 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	V1.0

A log event report signals new logbook entries.

Report

```
<frame>
 <report>
    <id> value_id </id>
    <ler>
        <log>
            <logEntry> value_logEntry </logEntry>
            <utcTime> value_utcTime </utcTime>
        </log>
        <log>      //opt
        ...
```

```
        </log>     //opt
    </ler>
  </report>
 </frame>
```

//opt ≙ Optional: Multiple logbook entries can be reported in a frame.

Reply

```
 <frame>
  <reply>
     <id> value_id </id>
     <resultCode> 0 </resultCode>
     <ler/>
  </reply>
 </frame>
```

A reply frame from the user application is optional. It is only necessary when transacted mode is set.

No negative replies are defined for reports.

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_logEntry	Text	--	Logbook entry
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.

3.4.1.5 presenceEventReport

Table 3-74 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	V1.0

A presence event report signals the number of transponders currently present in the antenna field whenever the number of transponders in the antenna field changes.

Report

```
<frame>
 <report>
    <id> value_id </id>
    <per>
        <presence>
           <sourceName> value_sourceName </sourceName>
           <noOfTags> value_noOfTags </noOfTags>
           <utcTime> value_utcTime </utcTime>
       </presence>
       ...
       <presence>       //opt
          ...
       </presence>      //opt
    </per>
 </report>
</frame>
```

//opt ≙ Optional: The presence can be signaled for multiple read points in a frame.

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <per/>
 </reply>
</frame>
```

A reply frame from the user application is optional. It is only necessary when transacted mode is set.

No negative replies are defined for reports.

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").

Parameter	Type	Values	Description
value_noOfTags	Decimal value	0...x	Number of transponders currently located in the antenna field. 0: No transponder in the antenna field 1: At least one transponder in the antenna field x: If supported by the reader, the exact number of transponders is specified if there is more than one transponder in the antenna field.
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.

3.4.1.6 lastAccessEventReport

Table 3-75 Version dependencies of the command (as of protocol version and firmware versions)

XML API	RF61xR/ RF650R/ RF68xR	RF69xR	RF1xxC	RF12xC	RF360R	RF1100
V3.0	V4.0	V1.0	V1.3	V1.0	V1.0	--

A last access event report records the transponder data of the transponder that was read or written to with the last successful command. Using this report, all successfully executed transponder commands can be recorded - even those that are issued in parallel by a controller.

Report

```
<frame>
 <report>
    <id> value_id </id>
    <laer>
        <client>
            <clientName> value_clientName </clientName>
            <command> value_command </command>
            <tag>
                <tagID> value_tagID </tagID>
                <sourceName> value_sourceName </sourceName>     //opt
                <tagPC> value_tagPC </tagPC>                //opt
                <tagXPC> value_tagXPC </tagXPC>         //*
                <tagXPC2> value_tagXPC2 </tagXPC2>      //*
                <phase> value_phase </phase>            //*
                <phaseEnd> value_phaseEnd </phaseEnd>   //*
```

```
                <utcTime> value_utcTime </utcTime>          //opt
                <antennaName> value_antennaName </antennaName>  //opt
                <rSSI> value_rSSI </rSSI>                   //*
                <channel> value_channel </channel>          //opt
                <power> value_power </power>                //opt
                <polarization> value_polarization </polarization> //opt
                <tagField>                                 //opt
                   <bank> value_bank </bank>               //opt
                   <startAddress> value_startAddress </startAddress> //opt
                   <dataLength> value_dataLength </dataLength>   //opt
                   <data> value_data </data>
                </tagField>                                //opt
                <tagField>                                 //opt
                ...
                </tagField>                                //opt
            </tag>
            ...
            <tag>       //opt
               ...
            </tag>      //opt
        </client>
        <client>        //opt
        ...
        </client>      // opt
    </laer>
 </report>
</frame>
```

//opt ≙ Optional: The number of parameters varies depending on device configuration and/or device type.

//*: Depending on the Ident device used, the FW version or the WBM parameter assignment

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <laer/>
 </reply>
</frame>
```

A reply frame from the user application is optional. It is only necessary when transacted mode is set.

No negative replies are defined for reports.

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ clientName	Fixed values	XML1 XML2 XML3 XML4 WBM OPCUA EIP PNIO PB READPOINT	Name of the user application that executed the last successful transponder command. READPOINT: A device-internal application of the read point initiated the command.
value_command	Fixed values	See description	Name of the last successfully executed transponder command. The possible values depend on the user application used: • PNIO, PB, EIP (Ident profile): INVENTORY, WRITE_ID, LOCK_TAG_BANK, PHYSICAL_READ, PHYSICAL_WRITE, FORMAT, TAG_STATUS, GET, PUT, NEXT • PNIO, PB (FB 45): WRITE, READ, INIT, MDS-STATUS • OPCUA: Scan, ReadTag, WriteTag, KillTag, LockTag, SetTagPasword, Read, Write • WBM, XML: readTagIDs, writeTagID, readTagMemory, writeTagMemory, killTag, lockTagBank, getTagStatus • READPOINT: Observed
value_tagID	Hexadecimal value 0...9, A...F	--	UID or EPC ID in "RAW Hex Data Format" Example of a 96-bit EPC ID of an RF600 transponder: 3005FB63AC1F3681EC880468 For additional information, refer to the "EPC Global Specification".
value_rwData	Hexadecimal value 0...9, A...F	--	Read/written data of the last command Is only output if the last command was a transponder-specific read/write command.

Parameter	Type	Values	Description
value_ sourceName	Text	--	Read point name The name is specified in the WBM of the device (WBM menu "Read points").
value_tagPC	Hexadecimal value 0...9, A...F	--	Tag PC (Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC [1]	Hexadecimal value 0...9, A...F	--	Tag-XPC (extended Protocol Control) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_tagXPC2 [1]	Hexadecimal value 0...9, A...F	--	Tag-XPC2 (extended Protocol Control 2) 16-bit value represented by four hexadecimal characters. Example: The value "1234" corresponds to the binary value "0001.0010.0011.0100".
value_phase [1]	Decimal value 0...9	0...360	Phase information in degrees
value_phaseEnd [1]	Decimal value 0...9	0...360	Phase information at the end of the reading in degrees
value_utcTime	Time	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_ antennaName	Fixed values	Antenna01 ... Antenna08	Name of the antenna
value_rSSI [1]	Decimal value 0...9	0...255 [2] -10000...+10000 [3]	RSSI value This value has no unit and is not directly related to the performance. [2] RSSI values in 1/100 dBm [3]
value_channel	Decimal value 0...9	1...50	Channel number on which the transponder was detected.
value_power	Decimal value with two decimal places 0...9	0, 5.00...33.00	Used radiated power of the antenna in [dB] Increment: 0.25 dB
value_ polarization	Fixed values	Circular Vertical Horizontal Unknown	Polarization of the antenna With external antennas always unknown.

Parameter	Type	Values	Description
value_bank	--	0...3	Memory bank of the read/written data of the last successfully executed transponder command. Multiple memory banks are also possible with the "XML" user applications, depending on the command. RF600: • 0: Reserved • 1: EPC • 2: TID • 3: USER MEMORY RF300: 0
value_ startAddress	Decimal value 0...9	0...65535	Start address of the first byte of the read/written data of the last successfully executed transponder command for reading/writing. Multiple memory banks are also possible with the "XML" user applications, depending on the command.
value_ dataLength	Decimal value 0...9	1...1024	Number of bytes of the read/written data of the last successfully executed transponder command for reading/writing. Multiple memory banks are also possible with the "XML" user applications, depending on the command.
value_data	Hexadecimal value 0...9, A...F	--	Read/written data of the last successfully executed transponder command for reading/writing. Multiple memory banks are also possible with the "XML" user applications, depending on the command. Each byte is represented by two hexadecimal characters. Example: The byte sequence "0x12, 0x34, 0xA3" is represented as the character string "1234A3" in the "value_data" parameter. "value_dataLength" is 3 in this example. If the transponder is detected but the data cannot be read (e.g. if the transponder has no user memory as required), this field remains empty. "value_success" is then set to "False".

1) Depending on the Ident device used, the FW version or the WBM parameter assignment

2) Only for the RF61xR/RF650R/RF68xR readers

3) Only for the RF69xR readers

3.4.2 alarm

This section describes all alarms. Alarms are asynchronous frames and are used as status, warning or error messages of the reader. Note that specific errors are usually reported via the reply frames of the respective commands.

The meaning of an alarm is indicated by the error number ("value_errorNumber") and other optional parameters.

Alarm report

```
<frame>
 <alarm>
    <id> value_id </id>
    <error>
        <utcTime> value_utcTime </utcTime>
        <errorNumber> value_errorNumber </errorNumber>
        <errotText> value_errotText </errotText>
        <deviceName> value_deviceName </deviceName>     //opt1
        <commandName> value_commandName </commandName>  //opt1
        <clientName> value_clientName </clientName>     //opt1
        <eventType> value eventType </eventType>      //opt
        <paramXY> value_param_xy </paramXY>           //opt
        ...
    </error>
 </alarm>
</frame>
```

// opt ≙ Optional: The parameters may be present, depending on the alarm message.

- 1: For all Ident devices except the RF61xR/RF650R/RF68xR < V4.0 readers

Reply

```
<frame>
 <reply>
    <id> value_id </id>
    <resultCode> 0 </resultCode>
    <error/>
 </reply>
</frame>
```

A reply frame from the user application is optional. It is only necessary when secured mode is set.

No negative replies are defined for reports.

Parameter

Parameter	Type	Values	Description
value_id	Decimal value 0...9	0...4294967295	Unique command identifier
value_ errorNumber	Decimal value 0...9	0...65535	Error number You will find a list of error numbers in the following table.
value_utcTime	Text	--	UTC time stamp (Universal Time Coordinated) in ISO 8601 format: yyyy-MM-ddTHH:mm:ss.fffzzz e.g.: 2018-12-24T18:34:56.929+00:00.
value_errorText	Text	--	Message text of the alarm
value_ deviceName [1]	Fixed values	Device00 Device01 Device02 Device03 Device04	Name of the device that is connected to the respective interface. "Device00" refers here to the basic device (e.g. RF188C). "Device01"... "Device04" refer to all devices connected to the respective interface of the basic device.
value_ commandName [1]	Text	--	Name of the executed command
value_ clientName [1]	Fixed values	XML1 XML2 XML3 XML4 WBM OPCUA EIP PNIO PB READPOINT	Name of the user application that executed the command. READPOINT: A device-internal application of the read point initiated the command.
value_ eventType	Fixed values	Coming Going	Status display Some alarms have a simple status indication. This indicates whether the error occurred and is still present or whether it has been eliminated.
value_paramXY	Text	--	Additional parameter The name of the parameter depends on the error. Depending on the error number, an alarm contains a different number of parameters.

[1] For all Ident devices except the RF61xR/RF650R/RF68xR < V4.0 readers

List of possible alarm messages

Error number ("value_errorNumber")		Description
hex	dec	
0x1511	5393	ERROR_NO_ANSWER_FROM_TAG The transponder is not responding.
0x1512 [1)]	5394	ERROR_WRONG_PASSWORD The password you entered is incorrect. Access to the transponder was denied.
0x1513 [1)]	5395	ERROR_VERIFY_TAG_FAILED Verification of the transponder failed.
0x1514 [1)]	5396	ERROR_TAG_UNSPECIFIED General transponder error
0x1515 [1)]	5397	ERROR_TAG_INSUFFICIENT_POWER There is not enough transponder power.
0x1521	5409	ERROR_NO_TAG There is no transponder in the antenna field.
0x1522	5410	ERROR_NO_DATA The requested data is not available.
0x1523 [1)]	5411	ERROR_INVALID_CRC Bad checksum
0x1525 [1)]	5413	ERROR_NO_FREQUENCY No radio channel activated.
0x1526 [1)]	5414	ERROR_NO_CARRIER No carrier signal activated.
0x1527	5415	ERROR_MORE_THAN_ONE_TAG_IN_FIELD There is more than one transponder in the antenna field.
0x1528 [1)]	5416	ERROR_AIR_PROTOCOL_UNSPECIFIED General radio protocol error
0x154A	5450	General error
0x154D	5453	Internal firmware error
0x1567	5479	ERROR_READ_COMMAND_FAILED The read command failed.
0x1591 [1)]	5521	Antenna error at connector ANT 1
0x1592 [1)]	5522	Antenna error at connector ANT 2
0x1593 [1)]	5523	Antenna error at connector ANT 3
0x1594 [1)]	5524	Antenna error at connector ANT 4
0x1595 [2)]	5525	Antenna error at connector ANT 5
0x1596 [2)]	5526	Antenna error at connector ANT 6
0x1597 [2)]	5527	Antenna error at connector ANT 7
0x1598 [2)]	5528	Antenna error at connector ANT 8
0x1599 [1)]	5529	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_1 Wrong antenna type or wrong antenna version at connector ANT 1.

Error number ("value_errorNumber")		Description
hex	dec	
0x159A [1]	5530	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_2 Wrong antenna type or wrong antenna version at connector ANT 2.
0x159B [1]	5531	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_3 Wrong antenna type or wrong antenna version at connector ANT 3.
0x159C [2]	5532	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_4 Wrong antenna type or wrong antenna version at connector ANT 4.
0x159D [2]	5533	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_5 Wrong antenna type or wrong antenna version at connector ANT 5.
0x159E [2]	5534	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_6 Wrong antenna type or wrong antenna version at connector ANT 6.
0x159F [2]	5535	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_7 Wrong antenna type or wrong antenna version at connector ANT 7.
0x15A0 [2]	5536	ERROR_WRONG_TYPE_OR_VERSION_ANTENNA_8 Wrong antenna type or wrong antenna version at connector ANT 8.
0x7B71	31601	Overflow - "alarm": Send buffer for "alarm" is full. Alarm messages can be discarded until the next alarm is received.
0x7B73	31603	Overflow - "tagEventReport": Send buffer for "tagEventReport" is full. Data can be lost until the next "tagEventReport" is received.
0x9BFD	39933	ERROR_PARAMETER_INVALID_VALUE A parameter has an invalid value.
0x9CC5	40133	ERROR_ANTENNA An antenna error was detected when executing the command.
0x9CC7	40135	Power supply error The power supply is very close to the low limit.
0x9D8E	40334	ERROR_TOO_MANY_TAGS There are too many transponders in the antenna field.
0x9D8F	40335	ERROR_FIELD_DISTURBED Field error or send error
0x9DF1	40433	ERROR_MEMORY_OVERRUN Access outside the addressed memory area.
0x9DEA	40426	ERROR_TAG_LOCKED The transponder to which data is meant to be written or which should be deactivated is locked.
0x7A152	500050	Connection attempt failed.
0x7A153	500051	Configuration successfully loaded. This alarm message is always sent to all connected clients (RF600 and RF1xxC) and also sent on activation of the Tag monitor.
0x7A154	500052	Connection established.
0x7A155	500053	Connection interrupted.

[1] Only for the RF61xR/RF650R/RF68xR readers

[2] Only for the RF69xR readers

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